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United States Patent	12414772
Kind Code	B2
Date of Patent	September 16, 2025
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Percutaneous tissue anchor techniques

Abstract

System for use with tissue of a heart comprises a tissue anchor comprising an engaging head that has a coupling eyelet that defines an eyelet plane, and a helical tissue-coupling element, reversibly anchorable to the tissue, and having a central helix axis that lies on the eyelet plane. The system also comprises an anchor driver, having a distal portion that has a casing that defines a slot, the slot defining a slot plane, and dimensioned to receive at least part of the eyelet therein in a manner in which the eyelet plane is coplanar with the slot plane. The anchor driver also comprises a detent, configured to reversibly inhibit movement of the eyelet with respect to the slot. The anchor driver is configured to anchor the tissue anchor to the tissue by rotating the tissue anchor by applying a rotational force to the eyelet. Other embodiments are also provided.

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Appl. No.: 17/827646

Filed: May 27, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220280161 A1	Sep. 08, 2022

Related U.S. Application Data

continuation parent-doc US 16427061 20190530 US 11344310 child-doc US 17827646
continuation parent-doc US 14437062 US 10376266 20190813 WO PCT/IL2013/050861

20131023 child-doc US 16427061
us-provisional-application US 61820979 20130508
us-provisional-application US 61784042 20130314
us-provisional-application US 61745848 20121226
us-provisional-application US 61717303 20121023

Publication Classification

Int. Cl.: **A61B17/12** (20060101); **A61B17/04** (20060101); A61B17/00 (20060101); A61B17/29 (20060101); A61F2/24 (20060101)

U.S. Cl.:

CPC **A61B17/12009** (20130101); **A61B17/0401** (20130101); A61B2017/00477 (20130101); A61B2017/00681 (20130101); A61B2017/0409 (20130101); A61B2017/0411 (20130101); A61B2017/0441 (20130101); A61B2017/0464 (20130101); A61B2017/0496 (20130101); A61B2017/2937 (20130101); A61F2/2445 (20130101)

Field of Classification Search

CPC: A61B (17/12009); A61B (17/0401); A61B (2017/00681); A61B (2017/0441); A61B (2017/0464); A61B (2017/0496)

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8343173	12/2012	Starksen et al.	N/A	N/A
8343174	12/2012	Goldfarb et al.	N/A	N/A
8343213	12/2012	Salahieh et al.	N/A	N/A
8349002	12/2012	Milo	N/A	N/A

8353956	12/2012	Miller et al.	N/A	N/A
8357195	12/2012	Kuehn	N/A	N/A
8382829	12/2012	Call et al.	N/A	N/A
8388680	12/2012	Starksen et al.	N/A	N/A
8393517	12/2012	Milo	N/A	N/A
8419825	12/2012	Burgler et al.	N/A	N/A
8430926	12/2012	Kirson	N/A	N/A
8449573	12/2012	Chu	N/A	N/A
8449599	12/2012	Chau et al.	N/A	N/A
8454686	12/2012	Alkhatib	N/A	N/A
8460370	12/2012	Zakay	N/A	N/A
8460371	12/2012	Hlavka et al.	N/A	N/A
8475491	12/2012	Milo	N/A	N/A
8475525	12/2012	Maisano et al.	N/A	N/A
8480732	12/2012	Subramanian	N/A	N/A
8518107	12/2012	Tsukashima et al.	N/A	N/A
8523881	12/2012	Cabiri et al.	N/A	N/A
8523940	12/2012	Richardson et al.	N/A	N/A
8545553	12/2012	Zipory et al.	N/A	N/A
8551161	12/2012	Dolan	N/A	N/A
8585755	12/2012	Chau et al.	N/A	N/A
8591576	12/2012	Hasenkam et al.	N/A	N/A
8608797	12/2012	Gross et al.	N/A	N/A
8628569	12/2013	Benichou et al.	N/A	N/A
8628571	12/2013	Hacohen et al.	N/A	N/A
8641727	12/2013	Starksen et al.	N/A	N/A
8652202	12/2013	Alon et al.	N/A	N/A
8652203	12/2013	Quadri et al.	N/A	N/A
8679174	12/2013	Ottma et al.	N/A	N/A
8685086	12/2013	Navia et al.	N/A	N/A
8715342	12/2013	Zipory et al.	N/A	N/A
8728097	12/2013	Sugimoto et al.	N/A	N/A
8728155	12/2013	Montorfano et al.	N/A	N/A
8734467	12/2013	Miller et al.	N/A	N/A
8734699	12/2013	Heideman et al.	N/A	N/A
8740920	12/2013	Goldfarb et al.	N/A	N/A
8747463	12/2013	Fogarty et al.	N/A	N/A
8778021	12/2013	Cartledge	N/A	N/A
8784481	12/2013	Alkhatib et al.	N/A	N/A
8790367	12/2013	Nguyen et al.	N/A	N/A
8790394	12/2013	Miller et al.	N/A	N/A
8795298	12/2013	Hernlund et al.	N/A	N/A
8795355	12/2013	Alkhatib	N/A	N/A
8795356	12/2013	Quadri et al.	N/A	N/A
8795357	12/2013	Yohanan et al.	N/A	N/A
8808366	12/2013	Braido et al.	N/A	N/A
8808368	12/2013	Maisano et al.	N/A	N/A
8845717	12/2013	Khairkhahan et al.	N/A	N/A
8845723	12/2013	Spence et al.	N/A	N/A
8852261	12/2013	White	N/A	N/A

8852272	12/2012	Gross et al.	N/A	N/A
8858623	12/2013	Miller et al.	N/A	N/A
8864822	12/2013	Spence et al.	N/A	N/A
8870948	12/2013	Erzberger et al.	N/A	N/A
8870949	12/2013	Rowe	N/A	N/A
8888843	12/2013	Khairkhahan et al.	N/A	N/A
8889861	12/2013	Skead et al.	N/A	N/A
8894702	12/2013	Quadri et al.	N/A	N/A
8911461	12/2013	Traynor et al.	N/A	N/A
8911494	12/2013	Hammer et al.	N/A	N/A
8926696	12/2014	Cabiri et al.	N/A	N/A
8926697	12/2014	Gross et al.	N/A	N/A
8932343	12/2014	Alkhatib et al.	N/A	N/A
8932348	12/2014	Solem et al.	N/A	N/A
8940044	12/2014	Hammer et al.	N/A	N/A
8945211	12/2014	Sugimoto	N/A	N/A
8951285	12/2014	Sugimoto et al.	N/A	N/A
8951286	12/2014	Sugimoto et al.	N/A	N/A
8961595	12/2014	Alkhatib	N/A	N/A
8961602	12/2014	Kovach et al.	N/A	N/A
8979922	12/2014	Jayasinghe et al.	N/A	N/A
8992604	12/2014	Gross et al.	N/A	N/A
9005273	12/2014	Salahieh et al.	N/A	N/A
9011520	12/2014	Miller et al.	N/A	N/A
9011530	12/2014	Reich et al.	N/A	N/A
9023100	12/2014	Quadri et al.	N/A	N/A
9072603	12/2014	Tuval et al.	N/A	N/A
9107749	12/2014	Bobo et al.	N/A	N/A
9119719	12/2014	Zipory et al.	N/A	N/A
9125632	12/2014	Loulmet et al.	N/A	N/A
9125742	12/2014	Yoganathan et al.	N/A	N/A
9138316	12/2014	Bielefeld	N/A	N/A
9173646	12/2014	Fabro	N/A	N/A
9180005	12/2014	Lashinski et al.	N/A	N/A
9180007	12/2014	Reich et al.	N/A	N/A
9192472	12/2014	Gross et al.	N/A	N/A
9198756	12/2014	Aklog et al.	N/A	N/A
9226825	12/2015	Starksen et al.	N/A	N/A
9265608	12/2015	Miller et al.	N/A	N/A
9326857	12/2015	Cartledge et al.	N/A	N/A
9414921	12/2015	Miller et al.	N/A	N/A
9427316	12/2015	Schweich, Jr. et al.	N/A	N/A
9474606	12/2015	Zipory et al.	N/A	N/A
9526613	12/2015	Gross et al.	N/A	N/A
9561104	12/2016	Miller et al.	N/A	N/A
9579090	12/2016	Simms et al.	N/A	N/A
9693865	12/2016	Gilmore et al.	N/A	N/A
9724084	12/2016	Groothuis et al.	N/A	N/A
9730793	12/2016	Reich et al.	N/A	N/A
9788941	12/2016	Hacohen	N/A	N/A

9801720	12/2016	Gilmore et al.	N/A	N/A
9907547	12/2017	Gilmore et al.	N/A	N/A
9937042	12/2017	Cabiri et al.	N/A	N/A
10368852	12/2018	Gerhardt et al.	N/A	N/A
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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. Ser. No. 16/427,061 to Herman et al., filed May 30, 2019, which published as US 2019/0321049, which is a continuation of U.S. Ser. No. 14/437,062 to Herman et al., filed Apr. 20, 2015, which published as US 2015/0272586, and which is the US National Phase of International Patent Application PCT/IL2013/050861 to Herman et al., filed Oct. 23, 2013, which published as WO 2014/064695, and which claims priority from: US Provisional Patent Application 61/717,303 to Sheps et al., titled "Controlled steering functionality for implant-delivery tool," filed Oct. 23, 2012; US Provisional Patent Application 61/745,848 to Sheps et al., titled "Controlled steering functionality for implant-delivery tool," filed Dec. 26, 2012; US Provisional Patent Application 61/784,042 to Herman et al., titled "Percutaneous tissue anchor techniques," filed Mar. 14, 2013; and US Provisional Patent Application 61/820,979 to Sheps et al., titled "Controlled steering functionality for implant-delivery tool," filed May 8, 2013, all of which are incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present application relates in general to medical devices. More specifically, the present application relates to tissue anchors and tool for manipulation of tissue anchors, such as for percutaneous use.

BACKGROUND

(2) Tissue anchors are used to facilitate implantation of medical implants, such as by coupling such implants to tissue. A tissue anchor may be driven through the implant and into the tissue while the implant is held in place. Alternatively, the tissue anchor may be implanted before introduction of the medical implant, and the medical implant subsequently coupled to the tissue anchor.

SUMMARY OF THE INVENTION

(3) Tissue anchor apparatus, and techniques for use thereof, are described. For some applications, a tool is described for manipulating tissue anchors. For some applications, the tool is configured for retrieval of already-implanted tissue anchors. For some applications, the tool is configured to be coupled to a tissue anchor at a non-zero angle with respect to the tissue anchor, and to subsequently be aligned with the tissue anchor. For some applications, the tool is configured to be advanced through a sleeve of an implant that has been implanted using tissue anchors, and to de-anchor at least one of the tissue anchors, thereby de-anchoring at least part of the sleeve from the tissue.

(4) For some applications, a tissue anchor is described having a helical tissue-coupling element and an engaging head that comprises an eyelet. For some such applications, the tool is configured to be initially articulatably coupled to the tissue anchor (e.g., to the eyelet), and to be subsequently rigidly-coupled to the tissue anchor. For some such applications, the tool is configured to couple to the eyelet such that, together, the eyelet and an anchor-engaging element of the tool resemble links

in a chain.

(5) For some applications, an anchor driver is described for implanting the tissue anchor.

(6) There is therefore provided, in accordance with an application of the present invention, apparatus for use with a tissue anchor, the apparatus including: a first element including: a first arm; and a second arm, shaped to define a concavity, and movably coupled to the first arm such that: the concavity faces the first arm, and defines at least part of a space, in an open state of the first element, a gap is defined between a distal portion of the first arm and a distal portion of the second arm through which a portion of the tissue anchor is movable into the space, and in a closed state of the first element, the portion of the tissue anchor is not movable through the gap; and a second element, movable with respect to the first element such that, while (1) the portion of the tissue anchor is disposed within the space, and (2) the first element is in the closed state thereof: in a first position of the second element with respect to the first element, the portion of the tissue anchor is movable within the space, and in a second position of the second element with respect to the first element, the second element is positioned to apply torque to the tissue anchor by rotation of the second element.

(7) In an application, in the open state, the distal portion of at least the second arm is disposed at an angle between 60 degrees and 80 degrees with respect to a longitudinal axis of the second element.

(8) In an application, in the open state, the distal portion of at least the second arm protrudes away from the second portion and toward a central longitudinal axis of the apparatus.

(9) In an application, the first arm and the second arm are disposed on a plane, and are more resistant to bending in-plane than off-plane.

(10) In an application: the first arm is shaped to define another concavity that defines at least part of the space, and the first arm is coupled to the second arm such that the concavity defined by the first arm faces the concavity defined by the second arm.

In an application, the portion of the tissue anchor includes a bar that has a thickness, and the gap has a width that is at least 5 times greater than the thickness of the bar.

(11) In an application: the portion of the tissue anchor includes a bar that has a thickness, and in the first position of the second element with respect to the first element the first arm and the second arm shape the space to define an aperture having a width that is at least 3 times greater than the thickness of the bar.

(12) In an application: the portion of the tissue anchor includes a bar that has a transverse cross-sectional area, and in the first position of the second element with respect to the first element the first arm and the second arm shape the space to define an aperture defining an area that is at least 10 times greater than the transverse cross-sectional area of the bar.

(13) In an application, in the first position of the second element with respect to the first element, the first arm and the second arm shape the space to define an aperture having (1) a depth, and (2) a width that is greater than the depth of the aperture.

(14) In an application, the apparatus is configured such that, while (1) the portion of the tissue anchor is disposed within the space, and (2) the first element is in the closed state thereof: when the first element is moved proximally, the first element applies a proximally-directed pulling force to the tissue anchor via contact between the first element and the portion of the tissue anchor, and in the second position of the second element with respect to the first element, the second element is positioned to apply torque to the tissue anchor via contact between the first element and the tissue anchor.

(15) In an application: the second element includes a housing that defines a compartment, in the first position, at least part of the first element is disposed outside of the compartment, and the second element is movable from the first position to the second position such that at least part of the first element becomes disposed within the compartment.

(16) In an application, in the open state of the first element, the gap is at least 1.5 times as great as a transverse width of the compartment.

- (17) In an application, the housing is configured to receive the portion of the tissue anchor at any of a plurality of rotational positions of the portion of the tissue anchor with respect to the housing.
- (18) In an application, the housing is configured to receive the portion of the tissue anchor at a continuum of rotational positions of the portion of the tissue anchor with respect to the housing, the continuum of rotational positions spanning at least 90 degrees.
- (19) In an application: the portion of the tissue anchor includes a bar that defines at least part of an eyelet that defines a plane, the eyelet having a width on the plane, the compartment has a transverse width that is greater than the width of the eyelet, and the housing including one or more protrusions that protrude radially inward into the compartment such that a distance across the compartment between the protrusions is smaller than the width of the eyelet.
- (20) In an application: the apparatus further includes an extracorporeal controller including a handle and an adjuster, the controller configured to move the second element with respect to the first element, the adjuster is coupled to the housing via a tubular longitudinal member and is configured to reversibly move the at least part of the first element into and out of the compartment by reversibly moving the housing further and closer from the handle, and the apparatus further includes a second longitudinal member that is disposed through tubular longitudinal member, and the handle is coupled to the first element via the second longitudinal member, such that a distance along the longitudinal member between the handle and the housing is fixed.
- (21) In an application, the housing is configured to move the first element from the open state to the closed state by moving with respect to the first element.
- (22) In an application, the apparatus is configured such that, when the second element moves from the first position to the second position, the housing inhibits movement of the portion of the tissue anchor within the space.
- (23) In an application: the apparatus further includes: the tissue anchor; and a tool for use with the tissue anchor, a distal portion of the tool including the first element and the second element, while (1) the portion of the tissue anchor is disposed within the space, and (2) the first element is in the closed state thereof: in the first position of the second element with respect to the first element, the tissue anchor is in an articulatably-coupled state with respect to the tool, in which the distal portion of the tool is deflectable with respect to the tissue anchor, and in the second position of the second element with respect to the first element, the housing inhibits deflection of the distal portion of the tool with respect to the tissue anchor.
- (24) In an application, in the articulatably-coupled state, the distal portion of the tool is deflectable in at least 1 steradian with respect to the tissue anchor.
- (25) In an application, in the articulatably-coupled state, the distal portion of the tool is deflectable in at least 180 degrees with respect to the tissue anchor.
- (26) In an application, in the articulatably-coupled state, the tool is rotatable with respect to the tissue anchor.
- (27) In an application, the apparatus further includes (1) an implant including a sleeve, and (2) a tool for use with the tissue anchor, a distal portion of the tool including the first element and the second element, and: the sleeve: defines a lumen and a proximal opening that provides access to the lumen, is configured to be anchored to tissue of the subject by the tissue anchor in a manner in which (1) a tissue-engaging element of the tissue anchor is disposed in the tissue, and (2) the portion of the tissue anchor is disposed in the lumen, and the distal portion of the tool is: configured to be transluminally advanced to the implant, dimensioned to access the lumen via the proximal opening, and configured to be coupled to the portion of the tissue anchor while within the lumen.
- (28) In an application, in the open state of the first element, the gap between the distal portion of the first arm and the distal portion of the second arm is at least 20 percent as great as a transverse cross-sectional width of the lumen.
- (29) In an application: the tissue anchor is one of a plurality of tissue anchors, the apparatus is for

use with the plurality of tissue anchors, the sleeve has a longitudinal axis, and is configured to be anchored to the tissue of the subject by the plurality of tissue anchors at a respective plurality of longitudinal sites of the sleeve, and the distal portion of the tool is dimensioned to access the tissue anchor while at least another one of the plurality of tissue anchors is disposed between the tissue anchor and the proximal opening of the sleeve.

(30) In an application, the apparatus further includes the tissue anchor, and: the portion of the tissue anchor includes a bar that defines at least part of an eyelet that defines a plane, the tissue anchor has a longitudinal axis that lies on the plane, and the eyelet has a height along the longitudinal axis of the tissue anchor that is greater than a width of the eyelet, the width of the eyelet being on the plane and orthogonal to the longitudinal axis.

(31) In an application, the bar is shaped to define an arch portion of the eyelet, and the arch portion is generally parabolic.

(32) In an application, at least the first arm is curved.

(33) In an application: the apparatus further includes a tool having a distal end that includes the first element and the second element, and the at least first arm is generally sigmoid and has an inwardly-convex proximal portion that curves away from a longitudinal axis of the distal end of the apparatus.

(34) In an application, the curvature of the at least first arm defines an inwardly-concave portion that defines the concavity.

(35) There is further provided, in accordance with an application of the present invention, apparatus, including: a tissue anchor including a tissue-engaging element and a head that includes a bar having a transverse cross-sectional area; and a tool including (1) a first part that includes an arm, and (2) a second part coupled to the first part such that: a space is defined between the first arm and the other part of the tool, in an open state, a gap is defined between the arm and the second part of the tool, the bar being movable through the gap and into the space, and in a closed state, the bar is not movable through the gap, and the space is shaped to define an aperture that has an area that is at least 10 times greater than the transverse cross-sectional area of the bar.

(36) There is further provided, in accordance with an application of the present invention, a method including: sliding, through a catheter, an anchor-manipulating tool, while the anchor-manipulating tool is coupled to a tissue anchor and articulatable with respect to the tissue anchor; inhibiting articulation between the anchor-manipulation tool and the tissue anchor; and while the articulation is inhibited, using the anchor-manipulating tool, performing an action selected from the group consisting of: driving the tissue anchor into tissue of a subject, and removing the tissue anchor from tissue of the subject.

(37) In an application, the step of inhibiting is performed subsequently to the step of sliding.

(38) In an application, the step of inhibiting is performed prior to the step of sliding.

(39) There is further provided, in accordance with an application of the present invention, apparatus for use with a tissue of a subject, the apparatus including: a tissue anchor, including an engaging head and a tissue-coupling element, reversibly anchorable to the tissue; and an anchor-retrieval tool, including: an anchor-engaging element including one or more arms, configured to be coupled to the engaging head, a housing that defines a compartment therein, and a controller, including an adjuster configured to move at least part of the anchor-engaging element into and out of the compartment.

(40) In an application, the adjuster is configured to move at least part of the anchor-engaging element into and out of the compartment by reversibly moving the housing, with respect to the anchor-engaging element along a longitudinal axis of the anchor-manipulation tool.

(41) In an application, the engaging head includes a bar that defines an eyelet that has a width, and the compartment has a transverse width that is greater than the width of the eyelet.

(42) In an application: the one or more arms include at least a first arm and a second arm that define a space therebetween, and the controller is configured to reversibly move the first arm and

the second arm between (1) a first state in which a gap between a distal portion of the first arm and a distal portion of the second arm is at least 5 times greater than a thickness of the bar, and (2) a second state in which the gap is smaller than the thickness of the bar, and the space between the first arm and the second arm defines an aperture that has a width that is at least 3 times greater than the thickness of the bar.

(43) In an application, a thickness of the one or more arms defines a depth of the aperture, and the width of the aperture is greater than the depth of the aperture.

(44) In an application, the anchor-retrieval tool has a distal end that includes the anchor-engaging element and the housing, and the apparatus has at least two states, including: an uncoupled state in which the anchor-engaging element is not in contact with the tissue anchor, and an articulatably-coupled state in which: the engaging head inhibits movement of the anchor-engaging element away from the tissue anchor, and the distal end of the anchor-retrieval tool is deflectable with respect to the tissue anchor,

the controller being configured to facilitate transition of the apparatus from the uncoupled state to the articulatably-coupled state.

(45) In an application, the engaging head includes an eyelet, and the controller is configured to facilitate transition of the apparatus from the uncoupled state to the articulatably-coupled state by moving at least part of the one or more arms through the eyelet

(46) In an application, the at least two states further include a rigidly-coupled state in which the distal end of the anchor-retrieval tool is inhibited from deflecting with respect to the tissue anchor.

(47) In an application, the adjuster is configured to move the apparatus from the articulatably-coupled state into the rigidly-coupled state by moving at least part of the engaging head into the compartment by moving the at least part of the anchor-engaging element into the compartment.

(48) In an application, when the apparatus is in the rigidly-coupled state, the anchor-manipulation tool is configured to apply a de-anchoring force the tissue anchor.

(49) In an application, the anchor-manipulation tool is configured to apply a rotational de-anchoring force to the anchor.

(50) In an application, the anchor-manipulation tool is configured to apply the rotational de-anchoring force by the housing rotating around a longitudinal axis of the anchor-manipulation tool.

(51) In an application, the one or more arms are curved.

(52) In an application, the one or more arms include exactly one arm.

(53) In an application, the exactly one arm is shaped to define a hook.

(54) In an application, the controller is configured to move the housing, with respect to the hook, along a longitudinal axis of the anchor-manipulation tool.

(55) There is further provided, in accordance with an application of the present invention, apparatus for use with a tissue of a subject, the apparatus including: a tissue anchor including an engaging head including an eyelet, and a tissue-coupling element reversibly anchorable to the tissue; and an anchor-manipulation tool: including, at a distal portion thereof, an anchor-engaging element including a curved arm, configured to be coupled to the tissue anchor by moving through the eyelet, and being configured, while the anchor-engaging element is coupled to the tissue anchor, to apply a rotational de-anchoring force to the tissue anchor.

(56) In an application, the curved arm is generally helical, and is configured to be coupled to the eyelet by being rotated with respect to the eyelet.

(57) In an application, the tissue-coupling element is generally helical and has a handedness, and the curved arm has a handedness that is opposite to the handedness of the tissue-coupling element.

(58) In an application, the anchor-engaging element includes at least a first arm and a second arm, the curve of each of the arms defining a respective concavity, and the arms positioned such that the concavity of the first arm faces the concavity of the second arm.

(59) In an application, the anchor-engaging element further includes a housing that defines a compartment therein, and an adjuster that is configured to move at least part of each of the arms

into and out of the compartment.

(60) In an application, when the anchor-engaging element is coupled to the tissue anchor, the anchor-manipulation tool is configured to move at least part of the eyelet into the compartment by the at least part of each of the arms moving into the compartment, and the housing is configured to apply the rotational de-anchoring force to the tissue anchor by the housing being rotated while the at least part of the eyelet is disposed within the compartment.

(61) In an application, the anchor-engaging element is configured to be coupled to the tissue anchor by being rotated with respect to the tissue anchor, and the anchor-manipulation tool is configured to apply the rotational de-anchoring force to the tissue anchor by at least part of the anchor-engaging element being rotated with respect to the tissue anchor.

(62) In an application, the anchor-engaging element further includes a housing that defines a compartment therein, and an adjuster that is configured to move at least part of the arm into and out of the compartment.

(63) In an application, when the anchor-engaging element is coupled to the tissue anchor, the anchor-manipulation tool is configured to move at least part of the eyelet into the compartment by the at least part of the arm moving into the compartment, and the housing is configured to apply the rotational de-anchoring force to the tissue anchor by the housing being rotated while the at least part of the eyelet is disposed within the compartment.

(64) In an application, the curved arm is generally helical, is configured to be coupled to the eyelet by being rotated with respect to the eyelet, and is configured to apply the rotational de-anchoring force to the tissue anchor by being further rotated with respect to the eyelet.

(65) There is further provided, in accordance with an application of the present invention, apparatus for use with a tissue of a subject, the apparatus including: a tissue anchor, including: an engaging head, including a coupling eyelet that defines a plane, and a helical tissue-coupling element, reversibly anchorable to the tissue, and having a central helix axis that lies on the plane defined by the eyelet; and an anchor driver, having a distal portion that is transluminally advanceable toward the tissue, and including: a casing that defines a slot dimensioned to receive at least part of the eyelet therein, and a detent, configured to reversibly inhibit movement of the eyelet with respect to the slot, the anchor driver being configured to anchor the tissue anchor to the tissue by rotating the tissue anchor by applying a rotational force to the eyelet.

(66) In an application, the slot is dimensioned to receive the at least part of the eyelet snugly.

(67) In an application, the anchor driver further includes a controller, configured to move the detent between (i) a closed state in which the at least part of the eyelet is inhibited from exiting the slot, and (ii) an open state in which the at least part of the eyelet is slidable out of the slot.

(68) In an application: the casing defines a recess, at least part of the detent moves into the recess when the detent moves into the open state thereof, and the controller includes a rod, configured to move the detent into the closed state thereof by the rod being slid into at least part of the recess.

(69) In an application, the apparatus further includes an anchor-retrieval tool, transluminally advanceable independently of the anchor driver.

(70) In an application, the anchor-retrieval tool includes: an anchor-engaging element including one or more arms, configured to be coupled to the engaging head, a housing that defines a compartment therein, and a retrieval-tool controller, including an adjuster configured to move at least part of the anchor-engaging element into and out of the compartment.

(71) In an application, the anchor-retrieval tool: includes, at a distal portion thereof, an anchor-engaging element including a curved arm, configured to be coupled to the tissue anchor by moving through the eyelet, and is configured, while the anchor-engaging element is coupled to the tissue anchor, to apply a rotational de-anchoring force to the tissue anchor.

(72) In an application: the anchor-retrieval tool includes: at a distal portion of the anchor-retrieval tool, an anchor-engaging element and an anchor-engaging-element actuator, and at a proximal portion of the anchor-retrieval tool, a retrieval-tool controller configured to control the anchor-

engaging-element actuator, the apparatus has at least two states: (1) an uncoupled state in which the anchor-engaging element is not in contact with the tissue anchor, and (2) an articulatably-coupled state in which: the engaging head at least in part inhibits movement of the anchor-engaging element away from the tissue anchor, and the distal portion of the anchor-retrieval tool is deflectable with respect to the tissue anchor, the anchor-engaging-element actuator is configured to facilitate transition of the apparatus from the uncoupled state to the articulatably-coupled state, and the anchor-retrieval tool, in at least one of the at least two states, being configured to apply a de-anchoring force the tissue anchor.

(73) There is further provided, in accordance with an application of the present invention, apparatus, including: an implant including a sleeve that includes a lateral wall that defines a lumen having a transverse cross-sectional diameter, and configured to be transluminally implanted in a subject; at least one tissue anchor including a coupling head and a tissue-engaging element, configured to be transluminally advanced to the implant, and to anchor the implant in the subject by the tissue-engaging element being driven through the lateral wall and into a tissue of the subject; and an anchor-retrieval tool including an anchor-engaging element including two arms, the anchor-engaging element configured: to be transluminally advanced to the implant and into the lumen of the implant, to be moved into a position in which a gap between respective distal portions of the two arms is at least 20 percent as great as the transverse cross-sectional diameter of the lumen, to be subsequently coupled to the coupling head, and to subsequently apply a de-anchoring force to the tissue anchor.

(74) In an application, the anchor-engaging element is configured to be moved into a position in which a gap between respective distal portions of the two arms is at least 50 percent as great as the transverse cross-sectional diameter of the lumen.

(75) In an application, the anchor-engaging element is configured to be moved into a position in which a gap between respective distal portions of the two arms is at least 80 percent as great as the transverse cross-sectional diameter of the lumen.

(76) There is further provided, in accordance with an application of the present invention, apparatus for use with a tissue of a subject, the apparatus including: a tissue anchor, including an engaging head and a tissue-coupling element, and reversibly anchorable to the tissue; and an anchor-manipulation tool including: at a distal portion of the tool, an anchor-engaging element and an anchor-engaging-element actuator, and at a proximal portion of the tool, a controller configured to control the anchor-engaging-element actuator,

the apparatus having at least two states, including: an uncoupled state in which the anchor-engaging element is not in contact with the tissue anchor, and an articulatably-coupled state in which: the engaging head at least in part inhibits movement of the anchor-engaging element away from the tissue anchor, and the distal portion of the anchor-manipulation tool is deflectable with respect to the tissue anchor, the anchor-engaging-element actuator being configured to facilitate transition of the apparatus from the uncoupled state to the articulatably-coupled state, and the anchor-manipulation tool, in at least one of the at least two states, being configured to apply a de-anchoring force the tissue anchor.

(77) In an application, in the articulatably-coupled state, the distal portion of the anchor-manipulation tool is rotatable with respect to the tissue anchor.

(78) In an application, the anchor-manipulation tool is configured to rotate the tissue anchor by applying the de-anchoring force to the tissue anchor.

(79) In an application, the at least two states further include a rigidly-coupled state in which the distal portion of the anchor-manipulation tool is inhibited from deflecting with respect to the tissue anchor.

(80) In an application, the anchor-manipulation tool is configured to apply the de-anchoring force to the tissue anchor while the apparatus is in the rigidly-coupled state.

- (81) In an application, the anchor-engaging-element actuator is configured to transition the apparatus from the articulatably-coupled state to the rigidly-coupled state.
- (82) In an application, the anchor-engaging-element actuator is configured to transition the apparatus from the articulatably-coupled state to the rigidly-coupled state by sliding, with respect to the anchor-engaging element, along a longitudinal axis of the distal portion of the tool.
- (83) In an application, in the articulatably-coupled state, the distal portion of the anchor-manipulation tool is deflectable in at least 1 steradian with respect to the tissue anchor.
- (84) In an application, in the articulatably-coupled state, the distal portion of the anchor-manipulation tool is deflectable in at least 3 steradians with respect to the tissue anchor.
- (85) In an application, in the articulatably-coupled state, the distal portion of the anchor-manipulation tool is deflectable in at least 7 steradians with respect to the tissue anchor.
- (86) There is further provided, in accordance with an application of the present invention, a method for use with a tissue anchor that is implanted in a tissue of a subject, the method including: advancing a tool toward the tissue anchor; coupling the tool to the tissue anchor such that a longitudinal axis of the tool is disposed at a nonzero angle with respect to a longitudinal axis of the tissue anchor; and subsequently aligning the longitudinal axis of the tool with the longitudinal axis of the tissue anchor.
- (87) In an application, at least a portion of the tissue anchor is disposed within a sleeve of an implant, and couples at least a portion of the implant to the tissue, and advancing the tool toward the tissue anchor includes advancing the tool at least partly through the sleeve of the implant.
- (88) In an application, aligning the longitudinal axis of the tool with the longitudinal axis of the tissue anchor includes aligning the longitudinal axis of the tool with the longitudinal axis of the tissue anchor by deforming the tissue of the subject.
- (89) In an application, deforming the tissue includes deforming the tissue by applying a force to the tissue anchor using the tool.
- (90) In an application, the tissue anchor includes an eyelet and the tool includes an anchor-engaging element and a housing that is slidable over the anchor-engaging element, and the method further includes facilitating aligning the longitudinal axis of the tool with the longitudinal axis of the tissue anchor by sliding the housing over at least part of the eyelet.
- (91) In an application, coupling the tool to the tissue anchor includes articulatably coupling the tool to the tissue anchor.
- (92) In an application, the method further includes, subsequently to aligning the longitudinal axis of the tool with the longitudinal axis of the tissue anchor, inhibiting articulation of the tissue anchor with respect to the tool.
- (93) In an application, inhibiting the articulation includes inhibiting the articulation by sliding a portion of the tool over a portion of the tissue anchor.
- (94) In an application, the tissue anchor includes an eyelet and the tool includes an anchor-engaging element and a housing that is slidable over the anchor-engaging element, and inhibiting the articulation includes inhibiting the articulation by sliding the housing over at least part of the eyelet.
- (95) In an application, the method further includes, subsequently to aligning the longitudinal axis of the tool with the longitudinal axis of the tissue anchor, applying a force to the anchor using the tool.
- (96) In an application, the method further includes de-anchoring the anchor from the tissue by applying the force to the anchor.
- (97) In an application, applying the force includes applying a rotational force to the anchor.
- (98) In an application, applying the rotational force to the anchor includes applying the rotational force to the anchor by rotating the tool, and the method further includes (1) after aligning the longitudinal axis of the tool with the longitudinal axis of the anchor, and (2) before applying the rotational force to the anchor, rotating the tool without applying the rotational force to the anchor.

(99) There is further provided, in accordance with an application of the present invention, apparatus for use with a tissue anchor, the tissue anchor including a tissue-coupling element and a bar that defines at least part of a coupling eyelet coupled to the tissue-coupling element, the coupling eyelet having a width, the apparatus including: a housing: having an inner surface that defines, within the housing, a compartment, the compartment having a transverse width that is greater than the width of the eyelet, including at least two protrusions that protrude radially inward from the inner surface into the compartment, such that a distance across the compartment between the protrusions is smaller than the width of the eyelet, and shaped to define, at a distal end of the housing, an opening into the compartment; and at least two arms including at least a first arm and a second arm, each of the arms defining a distal portion thereof, the arms defining a space therebetween, and being movable between: an extended position in which at least the distal portion of each arm is disposed outside the compartment, and the distal portion of the first arm and the distal portion of the second arm define a gap therebetween through which at least part of the bar is movable into the space, and a retracted position in which at least part of the distal portion of each arm is disposed inside the compartment, and the first arm and the second arm define a loop that shapes the space to define an aperture, the apparatus being configured such that as the arms move from the extended position toward the retracted position, the arms form the loop before reaching the retracted position.

(100) In an application, the first arm and the second arm are disposed on a plane, and are more resistant to bending in-plane than off-plane.

(101) In an application, the transverse width of the compartment is between 5 percent and 30 percent greater than the width of the eyelet.

(102) In an application, the apparatus is configured to be advanced percutaneously into a subject.

(103) In an application, in the extended position, the distal portion of each arm is disposed an angle between 60 degrees and 80 degrees with respect to the longitudinal axis.

(104) In an application, the compartment defined by the inner surface of the housing is generally cylindrical, and the transverse width of the compartment includes a transverse diameter of the compartment.

(105) In an application, the arms are configured to progressively form the loop responsively to being progressively moved into the compartment.

(106) In an application, the compartment is generally cylindrical, and the transverse width includes a transverse diameter of the generally cylindrical compartment.

(107) In an application, the tissue anchor has a longitudinal axis, the apparatus has a longitudinal axis, and the apparatus is configured to engage the coupling eyelet of the tissue anchor by the bar moving through the gap into the space, at any of at least 180 degrees of deflection of the longitudinal axis of the apparatus in a plane in which the longitudinal axis of the tissue anchor lies, and at any of at least 300 degrees of deflection of the longitudinal axis of the apparatus around the longitudinal axis of the tissue anchor.

(108) In an application, the housing is configured to receive the eyelet at any of a plurality of rotational positions of the eyelet with respect to the housing.

(109) In an application, the housing is configured to receive the eyelet at a continuum of rotational positions of the eyelet with respect to the housing.

(110) In an application: the at least two protrusions include a first protrusion and a second protrusion, the apparatus has a longitudinal axis, the first protrusion and the second protrusion define a transverse axis therebetween, the transverse axis being orthogonal to the longitudinal axis of the apparatus, the eyelet defines a plane, and the housing is configured to receive the eyelet at a continuum of rotational positions of the plane of the eyelet with respect to the transverse axis, the continuum of rotational positions spanning at least 90 degrees around the longitudinal axis of the apparatus.

(111) In an application, the housing is configured such that the continuum of rotational positions spans at least 150 degrees around the longitudinal axis of the apparatus.

(112) In an application, the housing is configured to receive the eyelet at one or more continua of rotational positions of the plane of the eyelet, that span a total of at least 270 degrees around the longitudinal axis of the apparatus, with respect to the transverse axis.

(113) In an application, the apparatus further includes an extracorporeal controller, including a handle and an adjuster, the adjuster being configured to move the arms between the extended position and the retracted position.

(114) In an application: the adjuster is coupled to the housing via a tubular longitudinal member, and is configured to reversibly move the arms between the extended position and the retracted position by reversibly moving the housing further and closer from the handle, and the handle is coupled to the arms via a longitudinal member that passes through the tubular longitudinal member, such that a distance along the longitudinal member between the handle and the housing is fixed.

(115) In an application, the handle is coupled to a proximal end of the longitudinal member that is exposed from a proximal end of the tubular longitudinal member.

(116) In an application, the extracorporeal controller includes an indicator that indicates at least a state of the adjuster in which (1) the arms define the loop, and (2) the arms are not in the retracted position.

(117) In an application, as the arms move from the extended position toward the retracted position, the arms deflect toward each other such that the distal portion of the first arm becomes closer to the distal portion of the second arm.

(118) In an application, the distal portion of each arm defines a respective beveled edge, and the arms are configured such that when the arms move into the retracted position, the beveled edge of first arm mates with the beveled edge of the second arm.

(119) In an application, a proximal portion of the first arm is coupled to a proximal portion of the second arm.

(120) In an application, the arms are biased toward deflecting away from each other.

(121) In an application, the proximal portion of the first arm is fixedly coupled to the proximal portion of the second arm, and each arm includes a resilient material, and is configured to progressively bend responsively to being progressively moved into the compartment.

(122) In an application, in the extended position, the gap between the distal portion of the first arm and the distal portion of the second arm is greater than the transverse width of the compartment.

(123) In an application, in the extended position, the gap between the distal portion of the first arm and the distal portion of the second arm is at least 1.5 times as great as the transverse width of the compartment.

(124) In an application, the apparatus includes the tissue anchor.

(125) In an application, the tissue anchor includes a base that couples the eyelet to the tissue-coupling element.

(126) In an application, the tissue anchor has a longitudinal axis, and the eyelet has a height along the longitudinal axis that is at least 80 percent as great as the width of the eyelet.

(127) In an application, the height of the eyelet is approximately equal to the width of the eyelet.

(128) In an application, the height of the eyelet is greater than the width of the eyelet.

(129) In an application, the height of the eyelet is at least 50 percent greater than the width of the eyelet.

(130) In an application, the bar is shaped to define an arch portion of the eyelet, a crest of the arch portion being further from the tissue-coupling element than is a base of the arch portion.

(131) In an application, the arch portion is generally parabolic.

(132) In an application, the tissue anchor has a longitudinal axis, the eyelet has a height along the longitudinal axis, and any portion of the bar that is parallel to the longitudinal axis has a height along the longitudinal axis that is less than 50 percent of the height of the eyelet.

(133) In an application, the arms are configured to be coupled to the coupling eyelet by trapping the at least part of the bar within the aperture by forming the loop while the at least part of the bar is

disposed in the space between the arms.

(134) In an application, the arms are configured to be articulatably coupled to the coupling eyelet.

(135) In an application, the apparatus is configured such that, while the arms are coupled to the eyelet, when the arms move into the retracted position, at least part of the eyelet moves into the compartment.

(136) In an application, the apparatus further includes the tissue anchor, the bar of the eyelet has a curvature, and the curvature of the bar of the eyelet facilitates the at least part of the eyelet moving into the compartment.

(137) In an application, the apparatus further includes an overtube, slidable over the housing and, while the at least part of the eyelet is disposed within the compartment, slidable over at least part of the tissue anchor.

(138) In an application, the apparatus is configured such that, when the at least part of the eyelet moves into the compartment, the housing inhibits articulation of the coupling eyelet with respect to the arms.

(139) In an application, the apparatus is configured such that, while the at least part of the eyelet is disposed within the compartment, rotation of the housing with respect to the anchor presses the protrusions against the bar of the eyelet.

(140) In an application, the apparatus is configured such that, while the at least part of the eyelet is disposed within the compartment, and the protrusions are pressed against the bar of the eyelet, further rotation of the housing with respect to the anchor rotates the anchor.

(141) In an application, the apparatus includes the tissue anchor, and the tissue-coupling element of the tissue anchor includes a helical tissue-coupling element.

(142) In an application, the apparatus is configured to facilitate screwing of the tissue-coupling element into tissue of a subject.

(143) In an application, the apparatus is configured to facilitate unscrewing of the tissue-coupling element from tissue of a subject.

(144) In an application, when the arms are coupled to the coupling eyelet, the loop and the eyelet resemble links in a chain.

(145) In an application: the arms each further define a proximal stem portion, a distal end of the proximal stem portion being coupled to the distal portion of the respective arm, the housing has a central longitudinal axis from a proximal end to the distal end thereof, in the extended position: the proximal stem portion of each arm protrudes distally away from the housing and outward from the longitudinal axis, and the distal portion of each arm protrudes from the distal end of the proximal stem portion and inward toward the longitudinal axis.

(146) In an application, in the extended position, the distal portion of each arm protrudes from the distal end of the proximal stem portion distally away from the housing.

(147) There is further provided, in accordance with an application of the present invention, apparatus, including: a tissue anchor, having a longitudinal axis and including an engaging head, shaped to define an opening, a tool having a longitudinal axis, and including: a plurality of arms including at least a first arm and a second arm, an arm actuator, configured to affect an angle of the arms with respect to the longitudinal axis of the tool by sliding along the longitudinal axis of the tool,

the apparatus having: an articulatably-coupled state in which at least one of the arms of the tool is hooked through the opening, and the longitudinal axis of the tool is movable with respect to the longitudinal axis of the anchor, and a rigidly-coupled state in which the arm actuator inhibits movement of the longitudinal axis of the tool with respect to the longitudinal axis of the anchor by pushing on the arms.

(148) In an application, the engaging head is shaped to define an opening that has a first end and a second end, and is narrower at the first end than at the second end.

(149) There is further provided, in accordance with an application of the present invention, a

method for use with a tissue anchor that is implanted in a tissue of a subject, the method including: advancing a tool toward the tissue anchor; articulatably coupling the tool to the tissue anchor; subsequently changing an alignment of a longitudinal axis of the tool with respect to a longitudinal axis of the tissue anchor; and subsequently rigidly coupling the tool to the tissue anchor.

(150) There is further provided, in accordance with an application of the present invention, a tissue anchor for coupling to tissue of a subject using a tool, the tissue anchor including: a helical tissue-coupling element: configured to be coupled to the tissue by being screwed into the tissue, and having a longitudinal axis; and an eyelet: defining a plane, coupled to the helical tissue-coupling element such that the longitudinal axis lies on the plane, and configured to be reversibly coupled to the tool, to accept a rotational force applied by the tool, and to transfer the rotational force to the helical tissue-coupling element.

(151) There is further provided, in accordance with an application of the present invention, apparatus, including: an anchor-manipulation tool including a tissue-anchor-engaging element, the tissue-anchor-engaging element including: a plunger moveable proximally and distally along a central longitudinal axis of the tool, the plunger defining: a distal tapered force applicator having a proximal portion that is wider than a distal portion of the force applicator, and a proximal longitudinal element coupled to the proximal portion of the force applicator, the longitudinal element being narrower than the proximal portion of the force applicator; and one or more engaging structures disposed adjacently to the plunger, each of the one or more engaging structures including: a curved arm having an inner surface which defines a proximal portion which curves toward the longitudinal axis, and a central hook portion which curves away from the longitudinal axis, and a distal hook that is a moveable in response to movement of the curved arm, during a first period, the plunger is disposable in a proximal position in which the proximal portion of the force applicator pushes against the proximal portion of the curved arm in a manner in which the distal hook is maintained in a closed state, and during a second period, the plunger is slidable distally to a distal position in which the proximal portion of the force applicator is disposed adjacently to the central hook portion of the curved arm and the proximal portion of the curved arm inclines toward the longitudinal element in a manner in which the distal hook is moved away from the longitudinal axis by the curved arm to an opened state.

(152) In an application, the apparatus further includes a tissue anchor including an engaging head having a wall shaped so as to define an opening shaped so as to define a funnel having a wide proximal end and a narrow distal end, and: the engaging head is shaped to define an undercut at the narrow distal end, and the engaging head is shaped so as to define one or more recesses at the narrow distal end in a vicinity of the undercut.

(153) There is further provided, in accordance with an application of the present invention, a method, including: providing an anchor-manipulation tool including: a plunger including (1) a distal tapered force applicator having a proximal portion that is wider than a distal portion of the force applicator, and (2) a proximal longitudinal element coupled to the proximal portion of the force applicator, the longitudinal element being narrower than the proximal portion of the force applicator, and one or more engaging structures disposed adjacently to the plunger, each of the one or more engaging structures including: a curved arm having an inner surface which defines a proximal portion which curves toward the longitudinal axis, and a central hook portion which curves away from the longitudinal axis, and a distal hook that is moveable in response to movement of the curved arm, and, during the providing, the hook is in a closed state in which the plunger is disposed in a proximal position in which the proximal portion of the force applicator pushes against the proximal portion of the curved arm; and moving the distal hook away from the longitudinal axis by the curved arm to an opened state by pushing the plunger distally to a distal position in which the proximal portion of the force applicator is disposed adjacently to the central hook portion of the curved arm and the proximal portion of the curved arm inclines toward the longitudinal element.

(154) There is further provided, in accordance with an application of the present invention, apparatus, including: a tissue anchor including an engaging head having a wall shaped so as to define an opening shaped so as to define a funnel having a wide proximal end and a narrow distal end, the engaging head being shaped to define an undercut at the narrow distal end, the wall of the engaging head being shaped so as to define one or more grooves at the narrow distal end proximal to the undercut; an anchor manipulating tool having a tissue-anchor-engaging element configured to engage the wall of the engaging head, the tool having one or more engaging structures, each of the one or more engaging structures including: a curved arm having an inner surface which defines a proximal portion which curves toward the longitudinal axis, and a central hook portion which curves away from the longitudinal axis, and a distal hook that is moveable in response to movement of the curved arm; and an annular element coupled to the distal end of the tool configured to facilitate positioning of the tissue-anchor-engaging element with respect to the wall of the engaging head.

(155) The present invention will be more fully understood from the following detailed description of applications thereof, taken together with the drawings, in which:

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. 1A-B, 2A-B, and 3A-C are schematic illustrations of a tissue anchor and a tool for use with the tissue anchor, in accordance with some applications of the invention;
- (2) FIGS. 4A-E, 5A-B, and 6A-E are schematic illustrations of a system comprising the tissue anchor and the tool, and of use of the tool to retrieve the anchor from tissue of a subject, in accordance with some applications of the invention;
- (3) FIG. 7 is a schematic illustration of a proximal portion of the tool, including an extracorporeal controller, in accordance with some applications of the invention;
- (4) FIG. 8 is a schematic illustration of arms of the tool, in accordance with some applications of the invention;
- (5) FIGS. 9A-C are schematic illustrations of a procedure for implanting an annuloplasty ring structure to repair a mitral valve, in accordance with an application of the present invention;
- (6) FIGS. 10A-B are schematic illustrations of the tool being used to retrieve an anchor from within a lumen of an implant, in accordance with some applications of the invention;
- (7) FIGS. 11A-E are schematic illustrations of another tool and techniques for use with a tissue anchor, in accordance with some applications of the invention;
- (8) FIGS. 12A-E are schematic illustrations of another tool and techniques for use with a tissue anchor, in accordance with some applications of the invention;
- (9) FIGS. 13A-D are schematic illustrations of an anchor driver and techniques for use with a tissue anchor, in accordance with some applications of the invention;
- (10) FIGS. 14 and 15A-C are schematic illustrations of a system for engaging an already-implanted anchor and facilitating extraction (e.g., retrieval) of the anchor from tissue, in accordance with some applications of the invention;
- (11) FIGS. 16, 17, 18A-C, and 19 are schematic illustrations of a system for engaging an already-implanted anchor and facilitating extraction (e.g., retrieval) of the anchor from tissue, in accordance with some applications of the invention;
- (12) FIGS. 20A-C are schematic illustrations of a technique for use with a tool and a tissue anchor, in accordance with some applications of the invention; and
- (13) FIGS. 21A-C are schematic illustrations of a tool for use with a tissue anchor, in accordance with some applications of the invention.

DETAILED DESCRIPTION OF EMBODIMENTS

(14) Reference is made to FIGS. 1A-3C, which are schematic illustrations of a tissue anchor **40** and a tool **80** for use with the tissue anchor, in accordance with some applications of the invention. Anchor **40** is shown in two rotational views in FIGS. 1A-B, respectively. Anchor **40** comprises a tissue-coupling element **42**, and an engaging head comprising a coupling eyelet **46** defined by a bar (e.g., a shaft) **44**. Typically, tissue-coupling element **42** comprises a helical tissue-coupling element, configured to be screwed into, and optionally screwed out of, tissue of a subject. Eyelet **46** provides a coupling interface for tool **80** and/or another tool. As described hereinbelow, tool **80** is typically configured, and used, to retrieve anchor **40** from tissue of the subject (e.g., after the anchor has been previously implanted and/or used to facilitate implantation of an implant). For some applications, tool **80** is also configured and/or used to implant anchor **40**. For some applications, a different tool is used to implant anchor **40**.

(15) Anchor **40** has a central longitudinal axis **a1**, from a distal end of tissue-coupling element **42** to a proximal end of eyelet **46**. Typically, a central helix axis of helical tissue-coupling element **42** is coaxial with axis **a1**. Eyelet **46** typically defines a plane **a4** (see also FIG. 6A), on which the central helix axis and/or axis **a1** typically lie. Typically, a height **d1** of eyelet **46** is at least 80 percent as great as a width **d2** of the eyelet (e.g., approximately equal to width **d2**, or greater than width **d2**, such as at least 50 percent greater than width **d2**). Typically, bar **44** defines an arch portion of eyelet **46**, and anchor **40** (e.g., the engaging head thereof) comprises a base **48** that defines a base portion of the eyelet, and couples the eyelet to tissue-coupling element **42**. The arch portion of the eyelet is thereby oriented such that a crest **43** of the arch portion is further from tissue-coupling element **42** than is a base **45** of the arch portion. The shape (e.g., curvature) of the arch is typically such that the arch continues to widen for most, if not all, of its length from crest **43** to base **45**. Typically, any portion of bar **44** that is parallel to longitudinal axis **a1** has a height along the longitudinal axis that is less than 50 percent (e.g., less than 20 percent, such as less than 10 percent) of height **d1** of the eyelet. Such portions may be considered 'imposts' of the arch portion that are less than 50 percent (e.g., less than 20 percent, such as less than 10 percent) of the overall height of the arch portion. For some applications, the arch portion of the eyelet has a generally parabolic shape. Bar **44** has a thickness (e.g., a bore) **d3**.

(16) FIGS. 2A-B show side and distal-end views, respectively, of a distal portion of tool **80**. Tool **80** comprises a housing **82** that has an inner surface **84** that defines a generally cylindrical compartment **86** that has a transverse width (e.g., a diameter) **d4**. An opening **88** at a distal end **83** of the housing provides fluid communication between compartment **86** and the outside of the housing. Two or more protrusions **90** protrude radially inward from inner surface **84** into compartment **86**, such that a distance **d5** across the compartment between the protrusions is smaller than width **d4** of the compartment, and also smaller than width **d2** of eyelet **46** of anchor **40**. Width **d4** of compartment **86** is typically slightly greater than width **d2** of eyelet **46**, so as to accommodate the eyelet without allowing enough space for the eyelet to move significantly within the compartment. For example, width **d4** may be at least 5 percent, less than 30 percent, and/or between 5 and 30 percent greater (e.g., 10-20 percent greater) than width **d2**.

(17) Tool **80** further comprises an anchor-engaging element **91** comprising one or more (e.g., two) arms **92**, each arm defining a proximal stem portion **93** and distal portion **94** (e.g., a distal hook portion), and the arms typically curving inwardly toward each other so as to define a space **96** therebetween (e.g., between respective proximal stem portions). That is, the curve of each arm defines a concavity **103**, and the concavity of each arm faces the concavity of the other arm, so as to define at least part of space **96**. Alternatively, stem portion **93** may be generally straight. It is to be noted, that even for applications in which stem portion **93** is generally straight, arm **92** would typically still define a concavity at a site **104** where portion **93** meets distal portion **94**.

(18) FIGS. 2A-B (and FIG. 3A) show arms in an extended position thereof, in which at least the distal portion of each arm protrudes out of opening **88** and is disposed outside compartment **86**, and a gap between the distal portions has a width **d6** that is great enough for at least part of bar **44** to

move through, into space **96**.

(19) Typically, width **d6** is greater than thickness **d3** of bar **44** of eyelet **46** of anchor **40** (e.g., at least 5 times as great as thickness **d3**, such as at least 10 times as great as thickness **d3**, such as at least 20 times as great as thickness **d3**). Width **d6** is further typically greater than width **d4** of compartment **86**, e.g., at least 1.5 times as great, less than 4 times as great, and/or between 1.5 times and 4 times as great. Arms **92** (e.g., portion **93** and/or portion **94** thereof) have a thickness **d7** that is typically less than 0.5 times as great (e.g., less than 0.2 times as great, such as less than 0.1 times as great) as width **d6**, and/or less than width **d4** (e.g., less than 0.5 times as great as width **d4**, such as less than 0.2 times as great as width **d4**).

(20) For some applications, a plane **a5** on which arms **92** lie is typically disposed at least 20 degrees, up to 90 degrees, and/or between 20 and 90 degrees with respect to a transverse axis **a3** between protrusions **90**. For example, and as shown in FIG. 2B, plane **a5** may be disposed at about 45 degrees with respect to axis **a3**. This orientation is discussed hereinbelow with respect to FIGS. 6A-E.

(21) In the extended position, stem portion **93** of each arm **92** typically protrudes distally away from housing **82** and outward from a longitudinal axis **a2** of tool **80**, and distal portion **94** of each arm protrudes from a distal end of the stem portion and inward toward the longitudinal axis. Further typically, distal portion **94** of each arm protrudes from the distal end of stem portion **93** distally away from the housing. Distal portion **94** (e.g., an inner surface thereof) is disposed at an angle α_A with respect to longitudinal axis **a2**, angle α_A typically being greater than 40 degrees, less than 85 degrees, and/or between 40 and 85 degrees (e.g., between 50 and 80 degrees, such as between 50 and 70 degrees and/or between 70 and 80 degrees). Due at least in part to these geometries, arms **92** typically do not define effective hooks while in the extended position. It is hypothesized that such a configuration reduces a likelihood of arms **92** inadvertently hooking to tissue or apparatus within the body of the subject.

(22) Arms **92** further have a retracted position in which at least part of distal portion **94** of each arm is typically disposed inside compartment **86**, as shown in FIG. 3C. In the retracted position, arms **92** define a loop (e.g., distal portions **94** are in contact with each other), thereby shaping space **96** to define an aperture **97**.

(23) FIGS. 3A-C show stages in the transition of arms **92** from the extended position (FIG. 3A) to the retracted position (FIG. 3C). FIG. 3B shows an intermediate position of arms **92** that is partway between the extended position and the retracted position, illustrating that as the arms move from the extended position to the retracted position, the arms form the loop before assuming the retracted position (e.g., before arms **92** completely enter compartment **86**).

(24) As arms **92** move from the extended position toward the retracted position, the arms deflect toward each other such that distal portions **94** become closer to each other (e.g., such that a distance between the distal portions is smaller than thickness **d3** of bar **44**). Typically, distal portions **94** are at least 10 times further (e.g., at least 20 times further) from each other when in the extended position than when in the retracted position. Further typically, in at least the retracted position (and still further typically also in the intermediate position), the distal portion of one arm is in contact with the distal portion of another (e.g., the other) arm. For some applications, each distal portion is shaped to define a beveled edge, so as to facilitate contact (e.g., mating) of the distal portions. For example, when the arms move into the intermediate position and/or the retracted position, at least part of the beveled edge of one arm moves past at least part of the beveled edge of the other arm. The beveled edges and the resulting mating is shown most clearly in FIG. 3B.

(25) As arms **92** move toward the intermediate position and/or the retracted position, the angle at which distal portion **94** is disposed with respect to longitudinal axis **a2** approaches, and typically passes, 90 degrees, such that each arm forms a hook (e.g., as shown in FIGS. 3B-C). Typically, this angle at least reaches 90 degrees before distal portions **94** contact each other. It is hypothesized that such a configuration facilitates preliminary coupling (e.g., hooking) of a single arm **92** to eyelet **46**

before the arms reach the intermediate and/or retracted position. This may be advantageous, for example, when coupling tool **80** to a tissue anchor that is not disposed close to longitudinal axis **a2** of tool **80**.

(26) Tool **80** further comprises an extracorporeal controller **110**, which comprises a handle **112** and an adjuster **114**, such as a switch or a lever. Adjuster **114** is configured to move arms **92** between the extended and retracted states. As described with reference to FIGS. **4A-E**, arms **92** are typically moved between the extended and retracted states by moving housing **82** distally while the arms remain stationary (e.g., with respect to anchor **40** and/or tissue of the subject). Such a configuration facilitates coupling of arms **92** to eyelet **46** of the anchor with a reduced likelihood of the arms inadvertently moving away from the eyelet. As described in more detail hereinbelow with reference to FIG. **7**, typically (1) adjuster **114** is coupled to housing **82** and is configured to reversibly move the housing further and closer from handle **112**, and (2) handle **112** is coupled to arms **92** such that a distance along tool **80** between the handle and the housing is fixed.

(27) As shown in FIGS. **3A-C**, adjuster **114** is moved so as to move arms **92** between the extended position and the retracted position (e.g., via the intermediate position). Typically, and as described in more detail with reference to FIG. **7**, a tactile and/or visual indicator indicates at least a state of adjuster **114** in which arms **92** are in the intermediate position (as shown in FIG. **3B**). Further typically, controller **110** is configured to partially hold adjuster **114** in one or more states, such as in the state in which arms **92** are in the intermediate position.

(28) Reference is made to FIGS. **4A-E**, **5A-B**, and **6A-E**, which are schematic illustrations of a system **100**, comprising tissue anchor **40** and tool **80**, and of use of the tool to retrieve the anchor from tissue **10** of the subject, in accordance with some applications of the invention. FIG. **4A** shows anchor **40** having been previously implanted in tissue **10** (e.g., to facilitate implantation of an implant; not shown in FIG. **4A**, but a non-limiting illustrative example is described with reference to FIGS. **9A-10**). Tool **80** is advanced toward anchor **40** with arms **92** in the extended position, such that arms **92** pass around bar **44** of eyelet **46** (i.e., such that the bar passes through the gap between the distal portions of the arms and into the space between the arms). Housing **82** is moved distally over at least part of arms **92** (e.g., such that the housing presses on a housing-contacting surface **105** of the arms), such that the arms move into the intermediate position, trapping bar **44** within aperture **97** defined by the loop formed by the arms (FIG. **4B**). That is, aperture **97** acts as or defines an eyelet-retention area. It is to be noted that in this state (1) arms **92** are articulatably coupled to eyelet **46**, (2) the arms (e.g., the loop defined thereby) and eyelet **46** generally resemble links in a chain (e.g., this state is an articulatably-coupled state), and (3) eyelet **46** inhibits movement of tool **80** (e.g., anchor-engaging element **91** thereof) away from tissue anchor **40**.

(29) Typically, aperture **97** (1) defines an area that is at least 10 times greater (e.g., at least 20 times greater, such as at least 40 times greater) than that of a transverse cross-sectional area of bar **44** of anchor **40**, (2) has a width **d8** (see FIG. **3B**; e.g., a smallest width and/or a width transverse to longitudinal axis **a2**) that is at least 3 times greater (e.g., at least 5 times greater, such as at least 8 times greater) than thickness **d3** of bar **44**, and/or (3) has a width **d8** that is greater (e.g., at least 2 times greater e.g., at least 4 times greater, such as at least 10 times greater) than a depth of the aperture, which is defined by thickness **d7** of arms **92**. That is, aperture **97** is typically (1) wider than bar **44** (e.g., at least 3 times wider, e.g., at least 5 times wider, such as at least 8 times wider), and/or (2) wider than itself is deep (e.g., at least 2 times wider, e.g., at least 4 times wider, such as at least 10 times greater). These relative dimensions of aperture **97** and bar **44** facilitate the articulatable coupling of tool **80** to eyelet **46**, and/or coupling of tool **80** to the eyelet at a wide range of possible angles of attack, e.g., as described hereinbelow.

(30) For some applications, tool **80** (e.g., anchor-engaging element **91** thereof) comprises only one arm, and bar **44** is trapped by the one arm, e.g., by being hooked by the one arm, and/or by being trapped between the one arm and inner surface **84** of compartment **86**. For some applications, tool

80 comprises two arms, but only one of the arms defines a concavity. For example, the other one of the arms may be straight, and the space between the arms is shaped to define an asymmetric aperture when distal portion **94** of the arm that defines the concavity comes into contact with a distal portion of the straight arm (e.g., as described with reference to FIG. **21**, *mutatis mutandis*). (31) It is to be further noted that tool **80** is shown having been advanced toward, and coupled to eyelet **46** with a longitudinal axis **a2** of the tool (e.g., of housing **82**) at a nonzero angle of attack with respect to longitudinal axis **a1** of anchor **40**. FIG. **5A** more clearly shows that tool **80** is configured to engage (e.g., to be coupled to) eyelet **46** at a variety of angles of attack, including deflection in a plane defined by longitudinal axis **a1** of the tissue anchor, and around axis **a1**. For example, tool **80** may be coupled to eyelet **46** while axis **a2** is generally parallel to tissue **10**, perpendicular to the tissue, or at any angle of attack therebetween. That is, tool **80** is configured to engage the coupling eyelet at at least 180 degrees of deflection of longitudinal axis **a2** with respect to longitudinal axis **a1** (e.g., in a plane on which axis **a1** lies). Similarly, tool **80** may be coupled to eyelet **46** at at least most (e.g., all) rotational angles of attack around longitudinal axis **a1**, e.g., at at least 300 degrees of deflection of axis **a2** around axis **a1**. For some applications of the invention, this means that when tool **80** is advanced from a given direction (e.g., due to anatomical and/or other constraints), the tool is couplable to eyelet **46** at at least most (e.g., all) rotational angles of attack of anchor **40** around its longitudinal axis.

(32) The above possible angles of attack of tool **80** may also be translated into three-dimensional terms. For example, the possible angles of attack of tool **80** typically, together, define a three-dimensional angular span of at least 1 steradian (e.g., at least 3 steradians, such as at least 7 steradians) around eyelet **46**. Such a span is illustrated schematically, by way of example, as span **102** in FIG. **5B**.

(33) While system **100** is in the articulatably-coupled state (e.g., while tool **80** is articulatably coupled to anchor **40**), tool **80** is deflectable with respect to anchor **40**. Typically, in this state, tool **80** is deflectable into any of the angles described hereinabove as angles of attack. For example, tool **80** is typically deflectable into any angle that lies within the three-dimensional angular span(s) that are described with reference to FIGS. **5A-B**, such as span **102** (e.g., a three-dimensional angular span that is at least 1 steradian, such as at least 3 steradians, such as at least 7 steradians).

(34) For some applications, such as when an angle between axes **a1** and **a2** is relatively large, tool **80** is subsequently manipulated so as to reduce the angle between axes **a1** and **a2**, e.g., to at least in part align the tool and anchor **40** (FIG. **4C**). For example, tool **80** may be rotated, pulled slightly proximally (e.g., along axis **a2**) and/or moved orthogonally with respect to axis **a2**. Typically, tissue **10** is sufficiently soft and/or flexible that the tissue responsively deforms (e.g., temporarily), allowing anchor **40** to move into a position in which axes **a1** and **a2** are more closely aligned.

(35) Subsequently, arms **92** are moved into the retracted position, typically by housing **82** being slid over the arms (e.g., while arms **92** remain stationary), and thereby also over at least part of (e.g., most of, or all of) eyelet **46** (FIG. **4D**). Housing **82** thereby typically functions as an arm actuator. By moving arms **92** into the retracted position, at least part of eyelet **46** moves into the compartment of housing **82**. Although the retracted position is described hereinabove as having at least part of distal portion **94** of each arm typically disposed inside compartment **86**, as shown in FIG. **3C**, it is to be noted that for some applications, distal portions do not enter compartment **86**. For example, in the retracted position, distal portions **94** may remain just outside of compartment **86**, e.g., sufficiently close to the compartment that at least part of eyelet **46** moves into the compartment when the arms move into the retracted position.

(36) Housing **82** is configured to receive eyelet **46** at a plurality of rotational positions (e.g., a continuum of rotational positions) with respect to the housing. Portions of bar **44** of eyelet **46** become disposed in a circumferential space between protrusions **90**. For example, and as shown in FIG. **6A**, which is a cross-sectional view of system **100** with eyelet **46** disposed within compartment **86** of housing **82** (as indicated by section "VIA-VIB" in FIG. **4D**), transverse axis **a3**

between protrusions **90** of the housing may be disposed at a continuum of rotational positions with respect to plane **a4** defined by eyelet **46**. Typically, this continuum of rotational positions spans at least 90 degrees (e.g., at least 150 degrees, such as almost 180 degrees) around axis **a2**. For some applications, housing **82** is configured to receive eyelet **46** at one or more continua of rotational positions of plane **a4**, that span a total of at least 270 degrees around longitudinal axis **a2**. FIG. 6A schematically shows rotational positions in which eyelet **46** may be disposed, by way of “phantom” portions of bar **44**.

(37) Although as described in the above paragraph housing **82** is configured to receive eyelet **46** at a plurality (e.g., a continuum) of rotational positions with respect to the housing, the orientation of arms **92** (e.g., plane **a5** thereof) with respect to protrusions **90** (e.g., axis **a3** therebetween) may bias the eyelet (e.g., plane **a4** thereof) to be oriented in a particular rotational position with respect to the housing. For example, and as shown in FIG. 6A, the arrangement of protrusions may bias plane **a4** toward an angular span **99** that is generally centered orthogonally to axis **a5**. FIG. 6A schematically shows a rotational position of the eyelet (e.g., into which the eyelet is biased) by way of shaded portions of bar **44**, disposed generally central within angular span **99**. It is hypothesized that such biasing may inhibit axis **a4** from aligning with axis **a3** (which might otherwise inhibit entry of eyelet **46** into compartment **86** of housing **82**). It is further hypothesized that such biasing may advantageously align protrusions **90** and eyelet **46** in advance of rotation of anchor **40** (which is described hereinbelow with reference to FIGS. 4E and 6A-E). FIG. 6B shows, for clarity, the same arrangement of elements as FIG. 6A, but without the “phantom” portions of bar **44**, and with less annotation.

(38) As described hereinabove, in the intermediate position, arms **92** are articulatably coupled to eyelet **46**. The compartment of housing **82** is typically dimensioned such that the eyelet fits generally snugly therewithin. Thereby movement of eyelet **46** into the compartment of housing **82** typically inhibits articulation of the eyelet with respect to the arms. The state of system **100** shown in FIG. 4D (and FIG. 4E) may thereby, for some applications, be a rigidly-coupled state. Typically, the dimensions and/or shape of eyelet **46** (e.g., of bar **44** thereof), such as those described with reference to FIGS. 1A-B, facilitate entry of the eyelet into the compartment of housing **82**, such as by facilitating alignment of anchor **40** with tool **80**. For example, curvature of the arch portion of eyelet **46** may facilitate alignment of axes **a1** and **a2** as the eyelet is moved into the compartment of housing **82**.

(39) Subsequently, as shown in FIG. 4E, and as shown in cross section in FIGS. 6C-E, housing **82** is rotated so as to rotate, and thereby unscrew, anchor **40** from tissue **10**. Initially, rotation of housing **82** moves protrusions **90** until they engage (e.g., contact and press against) bar **44** of eyelet **46** (FIGS. 6C-D). Further rotation of housing **82** applies a de-anchoring rotational force to eyelet **46**, thereby rotating the eyelet and anchor **40** as a whole, and thereby unscrewing the anchor from tissue **10** (FIG. 6E). It is to be noted that, (1) whereas a proximal/pulling force is applied to anchor **40** by a first element of tool **80** (i.e., anchor-engaging element **91**, e.g., arms **92** thereof), a rotational/torque force is applied by a second element of the tool (i.e., housing **82**, e.g., protrusions **90** thereof). That is, pulling and rotation of anchor **40** are conducted by two distinct elements of tool **80** that are movable (e.g., longitudinally slidable) with respect to each other. It is to be further noted that rotating housing **82** in the reverse direction may screw anchor **40** into tissue **10**. For some applications, tool **80** may be used to re-anchor anchor **40** into tissue **10**, or may even be used for the initial anchoring of the anchor.

(40) It is to be noted that, although FIGS. 4A-5B show anchor **40** anchored alone to tissue **10**, anchor **40** is typically used to anchor another element, such as an implant, to the tissue. For example, and as shown in FIGS. 9A-10 mutatis mutandis, tissue-coupling element **42** may pierce a portion of an implant and penetrate tissue **10**, such that base **48** sandwiches the portion of the implant to the tissue.

(41) Reference is made to FIG. 7, which is a schematic illustration of a proximal portion of tool **80**,

including extracorporeal controller **110**, which comprises handle **112** and adjuster **114**, in accordance with some applications of the invention. As described briefly hereinabove with reference to FIGS. **3A-C**, typically (1) adjuster **114** is coupled to housing **82** and is configured to reversibly move the housing further and closer from handle **112**, and (2) handle **112** is coupled to arms **92** such that a distance along tool **80** between the handle and the housing is fixed. For some applications, adjuster **114** is coupled to a tubular longitudinal member **116** (e.g., via a cam **118**), the tubular longitudinal member being slidable with respect to handle **112**, and a distal end of the tubular longitudinal member being coupled to housing **82**. For some applications, the distal portion of tubular longitudinal member **116** comprises and/or defines housing **82**. For such applications, handle **112** is typically coupled (e.g., fixedly coupled) to a longitudinal member **120** (e.g., via a fastener **122**, such as a set screw), a distal end of the longitudinal member being coupled (e.g., fixedly coupled) to arms **92**, and tubular longitudinal member **116** is slidable over the longitudinal member. Typically, longitudinal member **120** is disposed through tubular longitudinal member **116**. Typically, and as shown in FIG. **7**, handle **112** is coupled to a proximal end of longitudinal member **120** that is exposed from a proximal end of tubular longitudinal member **116**.

(42) Tool **80** is typically used to retrieve anchor **40** percutaneously (e.g., transluminally). Housing **82**, tubular longitudinal member **116** and longitudinal member **120** are thereby configured to be percutaneously advanced to a site at which anchor **40** is implanted. For example, when tool **80** is configured to transluminally retrieve anchor **40**, housing **82**, tubular longitudinal member **116** and longitudinal member **120** are dimensioned to fit within the blood vessel(s) through which they are to be advanced, and the tubular longitudinal member and the longitudinal member are sufficiently flexible to follow the transluminal route.

(43) Typically, extracorporeal controller **110** comprises and/or defines an indicator **124** that indicates a state of adjuster **114**, and thereby the position of arms **92**. Indicator **124** may comprise a visual indicator, such as markings, and/or may comprise a tactile indicator, such as graduated ridges **126** with which a part of adjuster **114** interfaces. Indicator **124** typically indicates at least a state of adjuster **114** in which arms **92** are in the intermediate position in which they define the loop but are not in the retracted position. Indicator **124** may also indicate one or more states of adjuster **114** in which arms **92** are between the extended position and the intermediate position (e.g., partially closed), e.g., so as to facilitate advancement of tool **80** through a narrow lumen. Indicator **124** (e.g., ridges **126** thereof) may be configured to partially hold adjuster **114** in one or more states, such as in the state in which arms **92** are in the intermediate position.

(44) Reference is again made to FIGS. **4A-E**, **5A-B**, and **6A-E**. For some applications, each arm **92** has a generally sigmoid shape, in which (1) an inwardly convex proximal portion **99** of the arm (e.g., an inwardly convex proximal portion of stem portion **93**) curves away from axis **a2**, and (2) an inwardly concave distal portion **101** of the arm (e.g., distal portion **94**, and/or an inwardly convex distal portion of stem portion **93**) curves at least in part back toward axis **a2**, and defines concavity **103** described hereinabove. It is hypothesized that this configuration facilitates at least some of the functionalities described herein. For example: The outward curvature of portion **99** may facilitate sliding housing **82** thereover, so as to move arms **92** into the intermediate position. That is, portion **99** may have a housing-contacting surface **105** on which housing **82** presses when the housing slides over portion **99**. For example, the outward curvature of portion **99** may maintain, during movement of housing **82** over arm **92**, a generally constant interface angle α_C between the housing and the part of the arm (e.g., the part of surface **105**) that is currently in contact with the housing. It is hypothesized that this thereby maintains a constant resistance to movement of housing **82** over arm **92**, thereby facilitating control by the operating physician. The outward curvature of portion **99** and/or the inward curvature of portion **101** may facilitate the arms forming the loop and aperture **97** before assuming the retracted position (e.g., before arms **92** completely enter compartment **86**).

(45) Reference is made to FIG. **8**, which is a schematic illustration of arms **92**, in accordance with

some applications of the invention. Typically, the proximal stem portion **93** of arms **92** are coupled to each other (e.g., fixedly coupled to each other, or articulatably coupled to each other). For some applications, and as shown in FIG. **8**, a forked member **140**, comprising a continuous piece of material, defines arms **92** (e.g., each tine of the forked member defining a respective arm). For some applications, arms **92** are biased toward deflecting away from each other, such as by forked member **140** comprising a resilient material. For some such applications, the resilient material allows arms **92** to progressively bend and/or deflect toward each other as the arms are progressively moved into compartment **86** of housing **82** (e.g., as the arms progressively move toward the intermediate and/or retracted position thereof).

(46) Arms **92** are typically disposed on a plane **a5** (e.g., when in a relaxed and/or unconstrained state). For some applications, arms **92** are more resistant to bending in plane **a5** (i.e., in-plane) than away from plane **a5** (i.e., off-plane). For some such applications, when forked member **140** defines arms **92** and comprises a resilient material, the material is more flexible off-plane than it is in-plane. FIG. **8** shows in phantom arms **92** being bent off-plane. It is hypothesized that this relatively high off-plane flexibility of arms **92** facilitates decoupling of arms **92** from an element to which they are coupled (e.g., to which they are inadvertently coupled), such as by moving (e.g., rotating) tool **80** in order to bend and thereby release the arm from the element.

(47) Reference is now made to FIGS. **9A-C**, which are schematic illustrations of a procedure for implanting an annuloplasty ring structure **222** to repair a mitral valve **230**, in accordance with an application of the present invention. Annuloplasty ring structure **222** is used to repair a dilated valve annulus of an atrioventricular valve, such as mitral valve **230**. For some applications, the annuloplasty ring is configured to be placed only partially around the valve annulus (e.g., to assume a C-shape), and, once anchored in place, to be contracted so as to circumferentially tighten the valve annulus. The annuloplasty ring comprises a flexible sleeve **228** and a plurality of anchors **40**. An anchor deployment manipulator **261**, comprising an anchor driver **236** and a deployment element **238** at a distal end of the anchor driver, is advanced into a lumen of sleeve **228**, and, from within the lumen, deploys the anchors through a wall of the sleeve and into cardiac tissue, thereby anchoring the sleeve around a portion of the valve annulus.

(48) For some applications, annuloplasty ring structure **222** is implanted, mutatis mutandis, using techniques described in U.S. application Ser. No. 12/437,103, filed May 7, 2009 which published as US 2010/0286767, and which issued as U.S. Pat. No. 8,715,342, and/or U.S. application Ser. No. 12/689,635, filed Jan. 19, 2010 which published as US 2010/0280604, and which issued as U.S. Pat. No. 8,545,553, both of which are assigned to the assignee of the present application and are incorporated herein by reference.

(49) For some applications, annuloplasty ring structure **222** is implanted, mutatis mutandis, using techniques described in US Provisional Application 61/717,303, which is assigned to the assignee of the present application and is incorporated herein by reference.

(50) Annuloplasty ring structure **222** comprises an adjusting mechanism **246**. The adjusting mechanism comprises a rotatable structure, such as a spool, arranged such that rotation of the rotatable structure contracts the implant structure. The implant further comprises a longitudinal member **286**, such as a wire, which is coupled to the adjusting mechanism. A rotation tool (not shown) is provided for rotating the rotatable structure. The tool is configured to be guided along (e.g., over, alongside, or through) longitudinal member **286** (i.e. a guide member), to engage the rotatable structure, and to rotate the rotatable structure in response to a rotational force applied to the tool.

(51) The procedure typically begins by advancing a semi-rigid guidewire into a right atrium **220** of the patient. The procedure is typically performed with the aid of imaging, such as fluoroscopy, transesophageal echo, and/or echocardiography. The guidewire provides a guide for the subsequent advancement of an outer steerable catheter **212** therealong and into the right atrium. Catheter **212** typically comprises a 14-24 F sheath, although the size may be selected as appropriate for a given

patient. Catheter **212** is advanced through vasculature into the right atrium using a suitable point of origin typically determined for a given patient. For example: catheter **212** may be introduced into the femoral vein of the patient, through an inferior vena cava **223**, into right atrium **220**, and into a left atrium **224** transseptally, typically through the fossa ovalis; catheter **212** may be introduced into the basilic vein, through the subclavian vein to the superior vena cava, into right atrium **220**, and into left atrium **224** transseptally, typically through the fossa ovalis; or catheter **212** may be introduced into the external jugular vein, through the subclavian vein to the superior vena cava, into right atrium **220**, and into left atrium **224** transseptally, typically through the fossa ovalis.

(52) For some applications of the present invention, catheter **212** is advanced through inferior vena cava **223** of the patient (as shown) and into right atrium **220** using a suitable point of origin typically determined for a given patient.

(53) Catheter **212** is advanced distally until the sheath reaches the interatrial septum, and a resilient needle and a dilator (not shown) are advanced through catheter **212** and into the heart. In order to advance catheter **212** transseptally into left atrium **224**, the dilator is advanced to the septum, and the needle is pushed from within the dilator and is allowed to puncture the septum to create an opening that facilitates passage of the dilator and subsequently catheter **212** therethrough and into the left atrium. The dilator is passed through the hole in the septum created by the needle.

Typically, the dilator is shaped to define a hollow shaft for passage along the needle, and the hollow shaft is shaped to define a tapered distal end. This tapered distal end is first advanced through the hole created by the needle. The hole is enlarged when the gradually increasing diameter of the distal end of the dilator is pushed through the hole in the septum. The advancement of catheter **212** through the septum and into the left atrium is followed by the extraction of the dilator and the needle from within catheter **212**.

(54) FIG. **9A** shows the distal portion of catheter **212** disposed within atrium **224**, and a second steerable catheter **214**, containing annuloplasty ring structure **222** at a distal end thereof, having been advanced through catheter **212** into left atrium **224**. As shown in FIG. **9A**, a distal end portion of catheter **214** extends beyond the distal end of catheter **212**. The distal end portion of catheter **212** is steerable in a first plane that is parallel to a plane of the annulus of mitral valve **230**. The distal end portion of catheter **214** is steerable toward the annulus of valve **230** along a plane that is perpendicular with respect to the steering plane of catheter **212** and that is perpendicular with respect to valve **230**. The juxtaposition of the two steering planes, along with the ability to slide catheter **214** distally and proximally through catheter **212**, allows structure **222**, and portions thereof, to be placed at any site on the annulus of valve **230**.

(55) As shown in FIG. **9A**, a distal end **251** of sleeve **228** is positioned against an annulus **240** of mitral valve **230**, e.g., in a vicinity of a left fibrous trigone **242**. (It is noted that for clarity of illustration, distal end **251** of sleeve **228** is shown schematically in the cross-sectional view of the heart, although left trigone **242** is in reality not located in the shown cross-sectional plane, but rather out of the page closer to the viewer.) Alternatively, distal end **251** is positioned in a vicinity of a right fibrous trigone **244** of the mitral valve (configuration not shown). Further alternatively, the distal end of the sleeve is not positioned in the vicinity of either of the trigones, but is instead positioned elsewhere in a vicinity of the mitral valve, such as in a vicinity of the anterior or posterior commissure. Once positioned at the desired site, deployment manipulator **261** deploys a first anchor **40a** through the wall of sleeve **228**. Typically, the anchor is deployed by penetrating the wall of the sleeve in a direction parallel to a central longitudinal axis of deployment manipulator **261**, or anchor driver **236**, and/or parallel to central longitudinal axis of the anchor, into cardiac tissue. Typically, base **48** of anchor **40** (shown in FIGS. **1A-B**) secures a portion of sleeve **228** by sandwiching the portion of the sleeve against the tissue to which the anchor is anchored.

(56) Typically, a channel **218** is provided disposed snugly within sleeve **228**, and anchor driver **236** delivers anchors **40** via the channel. Channel **218** is more rigid than sleeve **228**, and is progressively slid out of the sleeve as subsequent anchors **40** are delivered. Anchors **40** are

typically deployed from a distal end of manipulator **261** while the distal end is positioned such that a central longitudinal axis through the distal end of manipulator **61** forms an angle with a surface of the cardiac tissue of between about 20 and 90 degrees, e.g., between 45 and 90 degrees, such as between about 75 and 90 degrees, such as about 90 degrees. Typically, anchors **40** are deployed from the distal end of manipulator **261** into the cardiac tissue in a direction parallel to the central longitudinal axis through the distal end of manipulator **261**. Such an angle is typically provided and/or maintained by channel **218** being more rigid than sleeve **228**. The distal end of channel **218** is typically brought close to the surface of the cardiac tissue (and the wall of sleeve **228** that is disposed against the surface of the cardiac tissue), such that little of each anchor **40** is exposed from the channel before penetrating the sleeve and the tissue. For example, the distal end of channel **218** may be placed (e.g., pushed) against the wall of the sleeve, sandwiching the sleeve against the cardiac tissue.

(57) For some applications, this placement of the distal end of channel **218** against the cardiac tissue (via the wall of the sleeve), stabilizes the distal end during deployment and anchoring of each anchor **40**, and thereby facilitates anchoring. For some applications, pushing of the distal end of channel **218** against the cardiac tissue (via the wall of the sleeve) temporarily deforms the cardiac tissue at the site of contact. This deformation may facilitate identification of the site of contact using imaging techniques (e.g., by identifying a deformation in the border between cardiac tissue and blood), and thereby may facilitate correct positioning of the anchor.

(58) For some applications of the present invention, anchors **32** may be deployed from a lateral portion of manipulator **261**.

(59) As shown in FIG. **9B**, following the deployment of the first anchor, channel **218** is slid proximally out from a distal portion of sleeve **228**, typically facilitated by providing a reference pushing force to sleeve **228** using a reference-force tube (not shown). The distal end of channel **218** is repositioned along annulus **240** to another site selected for deployment of a second anchor **40b**, and the anchor is deployed.

(60) Typically, the first anchor is deployed most distally in the sleeve (generally at or within a few millimeters of the distal tip of the sleeve), and each subsequent anchor is deployed more proximally, such that the sleeve is gradually decoupled from channel **218** in a distal direction during the anchoring procedure (i.e., channel **218** is withdrawn from within sleeve **228**, so as to make the successive proximal portion sleeve **228** ready for implantation of a subsequent anchor). The already-deployed first anchor **40a** holds the anchored end of sleeve **228** in place, so that the sleeve is drawn from the site of the first anchor towards the site of the second anchor.

(61) As shown in FIG. **9C**, deployment manipulator **261** (e.g., deployment element **238** thereof) is repositioned along the annulus to additional sites, at which respective anchors (e.g., including anchors **40c** and **40d**) are deployed, until the last anchor is deployed in a vicinity of right fibrous trigone **244** (or left fibrous trigone **242** if the anchoring began at the right trigone). Alternatively, the last anchor is not deployed in the vicinity of a trigone, but is instead deployed elsewhere in a vicinity of the mitral valve, such as in a vicinity of the anterior or posterior commissure.

Longitudinal member **286** is typically left behind, and an adjustment tool **287** is subsequently threaded over and advanced along the guide member toward adjusting mechanism **246** and is used to rotate the spool of the adjusting mechanism, in order to tighten structure **222** by adjusting a degree of tension of a contracting member **226** disposed therewithin.

(62) FIGS. **9A-C** show longitudinal member **286**, coupled to adjusting mechanism **246**, and extending proximally through catheter **214** (e.g., into a lateral opening of the catheter, and through a secondary lumen of the catheter). It is to be noted that, for simplicity, rotation tool **287** and deployment element **238** are shown in FIG. **9C** as being simultaneously present in the heart of the subject. However, deployment element **238** may be withdrawn from the body of the subject before tool **287** is introduced. For some applications, catheter **214** is also removed, leaving longitudinal member **286** disposed through catheter **212**, such that adjustment tool **287** is advanceable through

catheter **212**, and along the guide member to adjusting mechanism **246**.

(63) Once the desired level of adjustment of structure **222** is achieved (e.g., by monitoring the extent of regurgitation of the valve under echocardiographic and/or fluoroscopic guidance), the rotation tool and longitudinal member **286** are removed from the heart. For some applications, a distal portion of longitudinal member **286** may be left within the heart of the patient and the proximal end may be accessible outside the body, e.g., using a port. For such applications, adjusting mechanism **246** may be accessed at a later stage following initial implantation and adjustment of ring structure **222**.

(64) For some applications, a re-access wire **288** may be provided, coupled to a proximal portion of the implant (e.g., a portion of the implant that is deployed last), such as to a last anchor (as shown in FIG. **9C**), or to the sleeve of the implant. Should it be determined, after implantation (e.g., and after adjustment) of annuloplasty ring structure **222**, that one or more anchors **40** requires retrieval, re-access wire **288** facilitates guidance of tool **80** to annuloplasty ring structure **222** and/or into the lumen thereof (described further with reference to FIGS. **10A-B**, *mutatis mutandis*).

(65) Alternatively, annuloplasty ring structure **222** is implanted by right or left thoracotomy, *mutatis mutandis*.

(66) Reference is made to FIGS. **10A-B**, which are schematic illustrations of tool **80** being used to retrieve an anchor **40** from within a lumen **164** of an implant **160**, in accordance with some applications of the invention. FIGS. **10A-B** show implant **160** comprising an annuloplasty ring structure **161** comprising a sleeve **162** that has been implanted by sequentially delivering a plurality of anchors **40** (e.g., at least anchors **40a**, **40b**, **40c**, and **40d**) (1) through lumen **164** defined by the sleeve, (2) through the sleeve wall and (3) into tissue **10**. Anchors **40** are typically delivered via a channel **166** that is disposed within sleeve **162** and is slidable progressively out of the sleeve as each anchor is delivered and anchored. For some applications, one or more elements shown in FIG. **10** comprise respective elements of the same name described with reference to FIGS. **9A-C**. For example, annuloplasty ring structure **161** may comprise annuloplasty ring structure **222**, sleeve **162** may comprise sleeve **228**, channel **166** may comprise channel **218**, and tissue **10** may comprise annulus **240**.

(67) Should it be necessary to remove an anchor, such as anchor **40d**, tool **80** may be advanced through channel **166** so as to access lumen **164**, and thereby the anchor (e.g., as shown in FIG. **10A**). Sleeve **162** typically facilitates this advancing by mechanically guiding (e.g., by partially constraining movement of) tool **80** and/or channel **166**. Typically, while tool **80** is being operated (e.g., while arms **92** are being moved between the extended and retracted positions), the arms are exposed out from channel **166** such that the channel does not inhibit movement of the arms. For example, while tool **80** is being operated, the arms may be disposed at least 5 mm, e.g., at least 10 mm, such as at least 50 mm distal to a distal end of channel **166**. Once anchor **40d** has been unscrewed from tissue **10** (and sleeve **162**), tool **80** is used to draw the anchor into channel **166** and out of the body of the subject. For some applications, tool **80** is advanced not via channel **166**, and the channel is subsequently advanced over the tool.

(68) For some applications, tool **80** may be guided to annuloplasty ring structure **161** and/or into lumen **164** thereof by being advanced along a re-access wire **288** (shown for structure **222** in FIG. **9C**, *mutatis mutandis*). Re-access wire **288** is coupled to a proximal portion of the implant (e.g., a portion of the implant that is deployed last), such as to a last anchor (as shown in FIG. **9C**), or to the sleeve of the implant.

(69) For some applications, should it be necessary to remove an anchor other than the most recently-deployed anchor, tool **80** may be coupled to that anchor by being advanced past more recently-deployed anchors. For example, as shown in FIG. **10B**, tool **80** may be advanced past anchors **40d** and **40c**, and coupled to anchor **40b**. Typically, but not necessarily, channel **166** is not advanced past the more recently-deployed anchors, but rather remains proximal to the most recently-deployed anchor. Lumen **164** of sleeve **162** has a transverse cross-sectional width (e.g.,

diameter) **d9**. Typically, tool **80** is dimensioned to pass, within the lumen, past the more recently-deployed anchors. Further typically, the flexibility of sleeve **162** facilitates such passage, e.g., by deforming in response to the presence of tool **80**.

(70) For some applications, flexibility of tubular longitudinal member **116** facilitates advancement of tool **80** through lumen **164**, e.g., past bends in sleeve **162**. For example, tool **80** may be advanced distally, and be at least in part guided and/or passively steered by sleeve **162** (e.g., tubular longitudinal member **116** may bend to conform with the direction of the sleeve). Such steering may be supplemented by rotation of tool **80**.

(71) While arms **92** of tool **80** are in the extended position, the gap between distal portions **94** of each arm (e.g., width **d6**; FIG. 2A) is typically at least 20 percent as great (e.g., at least 50 percent as great, such as at least 80 percent as great) as width **d9**. Such relative dimensions typically facilitate engagement of eyelet **46** by the arms, e.g., by increasing the likelihood of the eyelet becoming disposed between the arms. For some applications, the features of arms **92** (1) not defining effective hooks while in the extended position (described with reference to FIG. 2A) and/or (2) being more flexible off-plane than in-plane (described with reference to FIG. 8) facilitate use of such large relative dimensions because, as described hereinabove, these features are hypothesized to (1) reduce a likelihood of the arms inadvertently hooking to an untargeted tissue or apparatus (e.g., the sleeve, or another anchor), and/or (2) facilitate release of the arms from said untargeted tissue or apparatus. As described hereinabove, for some applications, tool **80** is also configured and/or used to implant anchor **40**. For some applications, tool **80** comprises and/or is analogous to deployment manipulator **261** and/or anchor driver **236** described with reference to FIGS. 9A-C (and arms **92** comprise and/or are analogous to deployment element **238**), *mutatis mutandis*.

(72) Imaging techniques such as fluoroscopy and/or ultrasound may be used to facilitate the use of tool **80**. For example, the position of arms **92** (including the rotational position) with respect to eyelet **46** may be observed using such imaging techniques, and adjusted accordingly. For some applications, fluoroscopy is used, following movement of arms **92** into the intermediate position, to identify successful coupling (e.g., articulatable coupling) of the arms to eyelet **46**, such as by identifying the links-in-a-chain arrangement described hereinabove. Similarly, imaging may be used to confirm that another element, such as sleeve **162**, has not been engaged by tool **80**. Following identification of the links-in-a-chain arrangement, the operating physician may then move the arms into the retracted position. For some applications, radiopaque markings are provided on arms **92**, eyelet **46**, and/or other components of the apparatus. For example, some elements of the apparatus may comprise tantalum, gold, platinum, or another radiopaque material. For some applications, mechanical guidance, such as that provided by sleeve **162** (as described hereinabove) is used to facilitate advancing of tool **80** (e.g., a distal end thereof) to the vicinity of the anchor, and imaging is used (e.g., only) to facilitate fine manipulation of the tool in order to couple the tool to the anchor.

(73) Throughout the present application various dimensions are mentioned, and the values thereof are typically described with respect to the values of other such dimensions. While not intending to unduly limit the scope of the invention, the following values, which have been determined to be of particular interest to the inventors, are provided:

(74) For some applications, height **d1** of eyelet **46** is greater than 1.5 mm and/or less than 7 mm, e.g., greater than 2 mm and/or less than 5 mm, such as greater than 2.3 mm and/or less than 2.6 mm (e.g., between 2.3 mm and 2.6 mm).

(75) For some applications, width **d2** of eyelet **46** is greater than 1.5 mm and/or less than 4 mm, e.g., greater than 2 mm and/or less than 3 mm, such as greater than 2.3 mm and/or less than 2.5 mm (e.g., between 2.3 mm and 2.5 mm).

(76) For some applications, thickness **d3** of bar **44** is greater than 0.15 mm and/or less than 1.5 mm, e.g., greater than 0.2 mm and/or less than 1 mm, such as greater than 0.25 mm and/or less than 0.4

mm (e.g., between 0.25 mm and 0.4 mm).

(77) For some applications, transverse width **d4** of compartment **86** is greater than 1.2 mm and/or less than 3.5 mm, e.g., greater than 1.5 mm and/or less than 2.6 mm, such as greater than 2 mm and/or less than 2.4 mm (e.g., between 2 mm and 2.4 mm).

(78) For some applications, distance **d5** between protrusions **90** is greater than 0.8 mm and/or less than 3 mm, e.g., greater than 1 mm and/or less than 2.2 mm, such as greater than 1.5 mm and/or less than 2 mm (e.g., between 1.5 mm and 2 mm).

(79) For some applications, width **d6** of the gap between distal portions **94** of arms **92** is greater than 2 mm and/or less than 9 mm, e.g., greater than 3 mm and/or less than 8 mm, such as greater than 5 mm and/or less than 7 mm (e.g., between 5 mm and 7 mm).

(80) For some applications, thickness **d7** of arms **92** is greater than 0.15 mm and/or less than 1.2 mm, e.g., greater than 0.2 mm and/or less than 0.8 mm, such as greater than 0.3 mm and/or less than 0.5 mm (e.g., between 0.3 mm and 0.5 mm).

(81) For some applications, width **d8** of the area defined by aperture **97** is greater than 1 mm and/or less than 3.5 mm, e.g., greater than 1.5 mm and/or less than 2.6 mm, such as greater than 2 mm and/or less than 2.4 mm (e.g., between 2 mm and 2.4 mm).

(82) For some applications, width **d9** of lumen **164** of sleeve **162** is greater than 1.8 mm and/or less than 9 mm, e.g., greater than 2.1 mm and/or less than 6 mm, such as greater than 2.5 mm and/or less than 4 mm (e.g., between 2.5 mm and 4 mm).

(83) Reference is made to FIGS. **11A-E**, which are schematic illustrations of a tool **380** and techniques for use with a tissue anchor such as tissue anchor **40**, in accordance with some applications of the invention. Similarly to tool **80**, tool **380** is also typically configured, and used, to retrieve anchor **40** from tissue of the subject (e.g., after the anchor has been previously implanted and/or used to facilitate implantation of an implant). Also similarly to tool **80**, may alternatively or additionally be configured and/or used to implant anchor **40**.

(84) Tool **380** comprises a housing **382** that has an inner surface **384** that defines a generally cylindrical compartment **386** that has a transverse width (e.g., a diameter), as shown in FIG. **2B** as width **d4** of compartment **86**, mutatis mutandis. An opening **388** at a distal end of the housing provides fluid communication between compartment **386** and the outside of the housing. For some applications, one or more protrusions **390** (seen in FIG. **11C**) protrude radially inward from inner surface **384** into compartment **386**. A distance across the compartment between the protrusions (as shown in FIG. **2B** as width **d5** of compartment **86**, mutatis mutandis) is smaller than the transverse width of the compartment, and also smaller than width **d2** of eyelet **46** of anchor **40**. The transverse width of compartment **386** is typically slightly greater than width **d2** of eyelet **46**, so as to accommodate the eyelet without allowing enough space for the eyelet to move significantly within the compartment. For example, the transverse width may be at least 5 percent, less than 30 percent, and/or between 5 and 30 percent greater (e.g., 10-20 percent greater) than width **d2**. Housing **382** is typically similar to, and for some applications is identical to, housing **82**, described hereinabove with reference to FIGS. **1A-10**.

(85) Tool **380** further comprises an anchor-engaging element **391** comprising at least one curved arm **392**. Typically, arm **392** is shaped to define a distal hook portion **394** which, for some applications is helical. Arm **392** is rotatable, independently of housing **382**, around a longitudinal axis **a6** of the housing (or of tool **380**) by rotating a longitudinal member **383** that extends from the arm out of the body of the subject. Arm **392** is also slidable, independently of housing **382**, along axis **a6** such that the arm (e.g., hook portion **394** thereof) is slidable into and out of the housing.

(86) FIG. **11A** shows arm **392** in an extended position thereof, in which at least hook portion **394** thereof protrudes out of opening **388** and is disposed outside compartment **386**. Arm **392** further has a retracted position in which at least part of hook portion **394** is typically disposed inside compartment **386**, as shown in FIG. **11C**.

(87) Tool **380** further comprises an extracorporeal controller (not shown), which comprises a

handle and an adjuster, such as a switch or a lever, coupled to housing **382** via a tubular longitudinal member **393**. For some applications, the controller of tool **380** is similar to, or comprises, controller **110** described hereinabove with reference to tool **80**. The adjuster is configured to move arm **392** between the extended and retracted states. As described for tool **80** (e.g., with reference to FIG. 7), mutatis mutandis, arm **392** is typically moved between the extended and retracted states by using tubular longitudinal member **393** to move housing **382** while the arm remains stationary (e.g., with respect to anchor **40** and/or tissue of the subject). Such a configuration facilitates coupling of arm **392** to eyelet **46** of the anchor with a reduced likelihood of the arm inadvertently moving away from the eyelet.

(88) FIG. **11A** shows anchor **40** having been previously implanted in tissue **10** (e.g., to facilitate implantation of an implant; not shown in FIG. **11A**, but a non-limiting illustrative example is described with reference to FIGS. **9A-10**). Tool **380** is advanced toward anchor **40** with arm **392** in the extended position (or the arm is moved into the extended position after advancing the tool toward the anchor).

(89) Arm **392** is rotated around axis **a6**, typically while housing **382** does not rotate. Hook portion **394** hooks through eyelet **46** of anchor **40**, preliminarily coupling tool **380** to the anchor (FIG. **11B**). In this state, tool **380** may be articulatably coupled to eyelet **46** (e.g., this state may be an articulatably-coupled state), e.g., as described hereinabove for tool **80**, mutatis mutandis. It is to be noted that in this state, eyelet **46** inhibits movement of tool **380** (e.g., anchor-engaging element **391** thereof) proximally away from tissue anchor **40**.

(90) For some applications of the invention, tool **380** (e.g., the extracorporeal controller thereof) is configured to detect resistance to rotation of arm **392**, and in response thereto, to provide an indication of this resistance, and/or to stop rotation of the arm. Typically, such resistance is due to successful hooking of eyelet **46**, and the indication and/or stopping of rotation may indicate to the operator to continue with the subsequent steps of the procedure. Alternatively, such resistance may be due to engagement of an undesirable structure, such as tissue of the subject or part of an implant, in which case the indication and/or stopping of rotation may indicate to the operator to reverse the rotation and/or to reposition arm **392**.

(91) It is to be noted that tool **380** is shown having been advanced toward, and coupled to eyelet **46** with a longitudinal axis **a6** of the tool (e.g., of housing **82**) at a nonzero angle of attack with respect to longitudinal axis **a1** of anchor **40**. Tool **380** is configured to engage (e.g., to be coupled to) eyelet **46** at a variety of angles of attack, including deflection in a plane defined by longitudinal axis **a1** of the tissue anchor, and around axis **a1**, e.g., as described for tool **80** with reference to FIGS. **5A-B**, mutatis mutandis.

(92) For example, tool **380** may be coupled to eyelet **46** while axis **a6** is generally parallel to tissue **10**, perpendicular to the tissue, or at any angle of attack therebetween. That is, tool **380** is configured to engage the coupling eyelet at at least 180 degrees of deflection of longitudinal axis **a6** with respect to longitudinal axis **a1** (e.g., in a plane on which axis **a1** lies). Similarly, tool **380** may be coupled to eyelet **46** at at least most (e.g., all) rotational angles of attack around longitudinal axis **a1**, e.g., at at least 300 degrees of deflection of axis **a6** around axis **a1**. For some applications of the invention, this means that when tool **380** is advanced from a given direction (e.g., due to anatomical and/or other constraints), the tool is couplable to eyelet **46** at at least most (e.g., all) rotational angles of attack of anchor **40** around its longitudinal axis.

(93) The above possible angles of attack of tool **380** may also be translated into three-dimensional terms. For example, the possible angles of attack of tool **380** typically, together, define a three-dimensional angular span of at least 1 steradian (e.g., at least 3 steradians, such as at least 7 steradians) around eyelet **46**.

(94) While tool **380** is articulatably coupled to anchor **40**, the tool is deflectable with respect to the anchor. Typically, in this state, tool **380** is deflectable into any of the angles described hereinabove as angles of attack.

(95) For some applications, such as when an angle between axes **a1** and **a6** is relatively large, tool **380** is subsequently manipulated so as to reduce the angle between axes **a1** and **a6**, e.g., to at least in part align the tool and anchor **40** (e.g., as described for tool **80** with reference to FIG. **4C**, *mutatis mutandis*). For example, tool **380** may be rotated, pulled slightly proximally (e.g., along axis **a6**) and/or moved orthogonally with respect to axis **a6**. Typically, tissue **10** is sufficiently soft and/or flexible that the tissue responsively deforms (e.g., temporarily), allowing anchor **40** to move into a position in which axes **a1** and **a6** are more closely aligned.

(96) Subsequently, arm **392** is moved into the retracted position, typically by housing **382** being slid over the arm (e.g., while the arm remains stationary), and thereby also over at least part of (e.g., most of, or all of) eyelet **46** (FIG. **11C**). Thus, by moving arm **392** into the retracted position, at least part of eyelet **46** moves into compartment **386**.

(97) Housing **382** is configured to receive eyelet **46** at a plurality of rotational positions (e.g., a continuum of rotational positions) with respect to the housing, such that portions of bar **44** of eyelet **46** become disposed in a circumferential space between protrusions **390**, e.g., as described hereinabove for housing **82**, *mutatis mutandis*.

(98) As described hereinabove, the preliminary coupling of arm **392** to eyelet **46** is an articulatable coupling. Compartment **386** is typically dimensioned such that the eyelet fits generally snugly therewithin. Thereby movement of eyelet **46** into compartment **386** typically inhibits articulation of the eyelet with respect to the arm. The state of tool **380** and anchor **40** shown in FIG. **11C** (and FIGS. **11D-E**) may thereby, for some applications, be a rigidly-coupled state. Typically, the dimensions and/or shape of eyelet **46** (e.g., of bar **44** thereof), such as those described with reference to FIGS. **1A-B**, facilitate entry of the eyelet into the compartment **386**, such as by facilitating alignment of anchor **40** with tool **380**. For example, curvature of the arch portion of eyelet **46** may facilitate alignment of axes **a1** and **a6** as the eyelet is moved into the compartment.

(99) Subsequently, as shown in FIGS. **11D-E**, housing **382** is rotated so as to rotate, and thereby unscrew, anchor **40** from tissue **10**. Initially, rotation of housing **382** moves protrusions **390** until they engage (e.g., contact and press against) bar **44** of eyelet **46** (e.g., as described for tool **80** with reference to FIGS. **6C-D**, *mutatis mutandis*). Further rotation of housing **382** applies a de-anchoring rotational force to eyelet **46**, thereby rotating the eyelet and anchor **40** as a whole, and thereby unscrewing the anchor from tissue **10** (e.g., as described for tool **80** with reference to FIG. **6E**, *mutatis mutandis*). It is to be noted that rotating housing **382** in the reverse direction may screw anchor **40** into tissue **10**. For some applications, tool **380** may be used to re-anchor anchor **40** into tissue **10**, or may even be used for the initial anchoring of the anchor.

(100) It is to be noted that, although FIGS. **11A-E** show anchor **40** anchored alone to tissue **10**, anchor **40** is typically used to anchor another element, such as an implant, to the tissue. For example, and as described hereinabove with reference to FIGS. **9A-10** *mutatis mutandis*, tissue-coupling element **42** may pierce a portion of an implant and penetrate tissue **10**, such that base **48** sandwiches the portion of the implant to the tissue. Similarly, tool **380** may be used to retrieve an anchor **40** from within a lumen of an implant, as described with reference to FIG. **10**, *mutatis mutandis*.

(101) Reference is made to FIGS. **12A-E**, which are schematic illustrations of a tool **480** and techniques for use with a tissue anchor such as tissue anchor **40**, in accordance with some applications of the invention. Similarly, to tool **80** and tool **380**, tool **480** is also typically configured, and used, to retrieve anchor **40** from tissue of the subject (e.g., after the anchor has been previously implanted and/or used to facilitate implantation of an implant). Tool **480** comprises an anchor-engaging element **491** comprising at least one curved arm **492**, typically coupled at a proximal end thereof to a mount **482**. Typically, arm **492** is generally helical, and is shaped to define a distal hook portion **494**. For some applications, portion **494** is simply the distal-most portion of the helix of arm **492**. For some applications, distal hook portion **494** has a smaller helix pitch than more proximal portions of the helix of arm **492**. For example, and as shown in the

“optional” bubble of FIG. 12D, the pitch may be close to zero, such that distal hook portion **494** is disposed generally orthogonal to axis **a7** of tool **480**, e.g., defining a planar arc.

(102) FIG. 12A shows anchor **40** having been previously implanted in tissue **10** (e.g., to facilitate implantation of an implant; not shown in FIG. 12A, but a non-limiting illustrative example is described with reference to FIGS. 9A-10). Tool **480** is advanced toward anchor **40**, and anchor-engaging element **491** (e.g., arm **492**) is rotated around a longitudinal axis **a7** of the tool (e.g., by rotating a longitudinal member **483** that extends out of the body of the subject). Hook portion **494** hooks through eyelet **46** of anchor **40**, preliminarily coupling tool **480** to the anchor (FIG. 12B). In this state, arm **492** may be articulatably coupled to eyelet **46** such (e.g., this state may be an articulatably-coupled state) such as described hereinabove for tool **80** and/or tool **380**, mutatis mutandis. It is to be noted that in this state, eyelet **46** inhibits movement of tool **480** (e.g., anchor-engaging element **491** thereof) proximally away from tissue anchor **40**.

(103) It is to be noted that tool **480** is shown having been advanced toward, and coupled to eyelet **46** with longitudinal axis **a7** of the tool at a nonzero angle of attack with respect to longitudinal axis **a1** of anchor **40**. Tool **480** is configured to engage (e.g., to be coupled to) eyelet **46** at a variety of angles of attack, including deflection in a plane defined by longitudinal axis **a1** of the tissue anchor, and around axis **a1**, e.g., as described for tool **80** with reference to FIGS. 5A-B, mutatis mutandis.

(104) For example, tool **480** may be coupled to eyelet **46** while axis **a7** is generally parallel to tissue **10**, perpendicular to the tissue, or at any angle of attack therebetween. That is, tool **480** is configured to engage the coupling eyelet at at least 180 degrees of deflection of longitudinal axis **a7** with respect to longitudinal axis **a1** (e.g., in a plane on which axis **a1** lies). Similarly, tool **480** may be coupled to eyelet **46** at at least most (e.g., all) rotational angles of attack around longitudinal axis **a1**, e.g., at at least 300 degrees of deflection of axis **a7** around axis **a1**. For some applications of the invention, this means that when tool **480** is advanced from a given direction (e.g., due to anatomical and/or other constraints), the tool is couplable to eyelet **46** at at least most (e.g., all) rotational angles of attack of anchor **40** around its longitudinal axis.

(105) The above possible angles of attack of tool **480** may also be translated into three-dimensional terms. For example, the possible angles of attack of tool **480** typically, together, define a three-dimensional angular span of at least 1 steradian (e.g., at least 3 steradians, such as at least 7 steradians) around eyelet **46**.

(106) While tool **480** is articulatably coupled to anchor **40**, the tool is deflectable with respect to the anchor. Typically, in this state, tool **480** is deflectable into any of the angles described hereinabove as angles of attack.

(107) At some point during rotation of tool **480**, anchor **40** provides resistance to further rotation of the tool. For example, hook portion **494** may abut base **48** of anchor **40**, and/or crest **43** of eyelet **46** of the anchor may abut a mount **482** to which anchor-engaging element **491** is coupled. For some applications of the invention, tool **480** (e.g., the extracorporeal controller thereof) is configured to detect resistance to rotation of anchor-engaging element **491** (e.g., arm **492**), and in response thereto, to provide an indication of this resistance, and/or to stop rotation of the arm (e.g., to provide the physician with an opportunity to decide whether to continue rotating the anchor-engaging element).

(108) Once this resistance is met, further rotation of anchor-engaging element **491** (e.g., arm **492**) applies a de-anchoring rotational force to eyelet **46**, thereby rotating the eyelet and anchor **40** as a whole, and thereby unscrewing the anchor from tissue **10** (FIGS. 12D-E). It is to be noted that the helix defined by tool **480** has an opposite handedness to the helix defined by tissue-coupling element **42** of anchor **40**. This allows continued rotation of anchor-engaging element **491** in one direction to (1) couple tool **480** to anchor **40**, and (2) unscrew the anchor from tissue **10**.

(109) For some applications, such as when an angle between axes **a1** and **a7** is relatively large, tool **480** is manipulated so as to reduce the angle between axes **a1** and **a7**, e.g., to at least in part align the tool and anchor **40** (e.g., as described for tool **80** with reference to FIG. 4C, mutatis mutandis).

For example, tool **480** may be rotated, pulled slightly proximally (e.g., along axis **a7**) and/or moved orthogonally with respect to axis **a7**. Typically, tissue **10** is sufficiently soft and/or flexible that the tissue responsively deforms (e.g., temporarily), allowing anchor **40** to move into a position in which axes **a1** and **a7** are more closely aligned (FIG. **12D**). Such manipulation of tool **480** may be performed before or after unscrewing of anchor **40** has begun.

(110) As described hereinabove, for some applications, distal hook portion **494** has a smaller helix pitch than more proximal portions of the helix of arm **492**. Such a configuration is hypothesized to facilitate alignment of tool **480** and anchor **40**, by becoming disposed flat against base **48** of the anchor (e.g., as shown in the “optional” bubble of FIG. **12D**). For such applications, portion **494** becoming disposed flat against base **48** inhibits articulation of tool **480** with respect to anchor **40**.

(111) Reference is made to FIGS. **13A-D**, which are schematic illustrations of an anchor driver **500** and techniques for use with a tissue anchor such as tissue anchor **40**, in accordance with some applications of the invention. Driver **500** comprises an anchor-engaging head **502** at a distal end of the driver, and a shaft **504** proximal to the anchor-engaging head. Shaft **504** is flexible, advanceable (e.g., transcatheterally) through vasculature of a subject, and typically has a length greater than 20 cm, and/or less than 2 m, such as greater than 50 cm and/or less than 1.5 m, e.g., between 90 cm and 1.2 m. For some applications, driver **500** comprises a handle **506** at a proximal end of shaft **504**, the handle comprising an adjuster **508** (e.g., a switch or a lever) configured to actuate engaging head **502**.

(112) Engaging head **502** is configured to be reversibly couplable to tissue anchor **40** (e.g., to the engaging head thereof), so as to facilitate driving of the anchor into tissue of the subject, and subsequent release of the anchor and withdrawal of driver **500** from the subject. Actuation of engaging head **502** by adjuster **508** typically transitions the engaging head between (i) a closed state in which the engaging head is coupled (e.g., locked) to anchor **40**, and (ii) an open state in which the engaging head is configured to release the anchor.

(113) Engaging head **502** comprises a casing **514** that defines a slot **520** and a recess **518** and further comprises a detent **510** that is reversibly movable (e.g., deflectable) within the casing, such as being movable into and out of the recess. Typically, (1) detent **510** is biased toward being disposed within recess **518**, such as by at least part of the detent (e.g., a stem portion **516**) comprising an elastically-deformable material, and (2) the detent is moved out of the recess by a controller, such as a rod **512**, applying a force to the detent, such as by being slid by adjuster **508** into at least part of the recess.

(114) Slot **520** is dimensioned to facilitate coupling of driver **500** to anchor **40** by receiving at least part of eyelet **46** of the anchor. For some applications of the invention, driver **500** is provided pre-coupled to anchor **40** (e.g., a kit is provided comprising a plurality of drivers, each driver coupled to a respective anchor). For some applications, driver **500** is coupled to anchor **40** by the operating physician or an assistant thereof.

(115) Movement of detent **510** out of recess **518** transitions engaging head **502** into a closed state thereof in which eyelet **46** is inhibited from exiting slot **520** and therefore in which anchor **40** is coupled to the engaging head. Movement of detent **510** into recess **518** transitions engaging head **502** into an open state thereof in which eyelet **46** is slidable out of slot **520** and thereby in which anchor **40** is decouplable from the engaging head.

(116) Driver **500** is advanced through vasculature of the subject while engaging head **502** is coupled to an anchor **40** as described hereinabove. Driver **500** is advanced toward tissue **10**, and the driver (e.g., at least the engaging head thereof) is rotated so as to screw tissue-coupling element **42** of anchor **40** into the tissue (FIG. **13A**). It is to be noted that, although FIGS. **13A-D** show driver **500** being used to anchor anchor **40** to tissue **10** alone, as described hereinabove (e.g., with reference to FIGS. with reference to FIGS. **4A-5B**, and **11A-E**, anchor **40** is typically used to anchor another element, such as an implant, to the tissue. For example, and as shown in FIGS. **9A-10** *mutatis mutandis*, tissue-coupling element **42** may pierce a portion of an implant and penetrate

tissue **10**, such that base **48** sandwiches the portion of the implant to the tissue. For some applications, anchor driver **500** comprises anchor driver **236**, described hereinabove with reference to FIGS. **9A-C**, and/or is used as described for anchor driver **236**, *mutatis mutandis*.

(117) Driver **500** (e.g., engaging head **502** thereof) typically rotates anchor **40** until base **48** of the anchor is firmly disposed against tissue **10** (FIG. **13B**), or against a portion of an implant that is being anchored to the tissue using the anchor. For example, base **48** may sandwich sleeve **228** and/or sleeve **162** described hereinabove against tissue **10** (e.g., as shown in FIGS. **9A-10**, *mutatis mutandis*).

(118) Subsequently, rod **512** is moved out of recess **518** (e.g., by being withdrawn proximally), and detent **510** responsively moves into the recess, thereby transitioning driver **500** (e.g., engaging head **502**) into the open state thereof (FIG. **13C**). Once in the open state, driver **500** is withdrawn proximally such that eyelet **46** slides out of slot **520**, leaving anchor **40** coupled to tissue **10** (FIG. **13D**).

(119) For some applications, anchor driver **500** is subsequently discarded, and any additional anchor is delivered and anchored using a respective additional anchor driver. For some applications, additional anchors are delivered and anchored using the same anchor driver.

(120) Reference is made to FIGS. **14-15C**, which are schematic illustrations of a system **2500** for engaging an already-implanted anchor and facilitating extraction (e.g., retrieval) of the anchor from tissue, in accordance with some applications of the invention. For some applications of the invention, system **2500** comprises elements of, and/or has similar features to, system **100**, described hereinabove. For some applications, system **2500** is used with techniques described in US Provisional Application 61/717,303, which is assigned to the assignee of the present application, and is incorporated herein by reference. System **2500** comprises an anchor-manipulation tool **2502**, comprising an anchor-engaging element **2504**, comprising two or more arms **2506** (e.g., arm **2506a** and arm **2506b**), and a housing **2508**. Arms **2506** are typically curved inwardly toward each other, as shown in FIGS. **35-37**.

(121) FIGS. **15A-C** show tool **2502** being coupled to a tissue anchor **2532**, e.g., prior to extraction of the anchor. Anchor **2532** comprises a tissue-coupling element **2536** (e.g., a helical tissue coupling element, as shown, or a screw), and coupling head **2534** which defines, or is coupled to, a coupling eyelet **2538**, which typically forms a closed loop. For some applications, tissue anchor **2532** comprises tissue anchor **40**, described hereinabove. For some applications, system **2500** comprises one or more anchors **2532** (e.g., tool **2502** and anchors **2532** are specifically manufactured to be used together).

(122) Tool **2502** is advanced toward anchor **2532** while anchor-engaging element **2504** is in an open position in which arms **2506** are held away from each other (FIG. **15A**), such that eyelet **2538** of the anchor moves between the arms. Subsequently, element **2504** is closed by moving arms **2506** toward each other. Typically, in the closed configuration, arms **2506** form a closed loop, such that element **2504** and eyelet **2538** generally resemble two links in a chain. For some applications, element **2504** is closed by moving housing **2508** distally, and thereby pushing arms **2506** together (FIG. **15B**). Alternatively or additionally, arms **2506** may be controlled via a control rod **2510** (shown in FIG. **14**).

(123) Housing **2508** is slid further distally, such that at least part of eyelet **2538** is disposed within the housing (FIG. **15C**). This “pushing” movement is typically facilitated by a counterforce (e.g., a “pulling” force) provided by element **2504** on anchor **2532**. Housing **2508** is shaped to define one or more (e.g., two) protrusions **2512**, which typically protrude from an inner surface of the housing, into a lumen of the housing. When eyelet **2538** is moved into the housing, portions of the eyelet are disposed in circumferential spaces between protrusions **2512**, as shown in state (i) of section A-A of FIG. **36C**. Housing **2508** is subsequently rotated, such that protrusions **2512** engage eyelet **2538**, as shown in state (ii) of section A-A of FIG. **36C**. In this state, further rotation of tool **2502** (e.g., simultaneous rotation of element **2504** and housing **2508**) rotates anchor **2532**, and thereby

facilitates extraction of the anchor from the cardiac tissue.

(124) System **2500** is typically used in a similar manner to that in which system **100** is used. For some applications, system **2500** is used as described for system **100** with reference to FIG. **10**, *mutatis mutandis*.

(125) Reference is again made to FIGS. **14-15C**. It should be noted that tool **2502** may engage eyelet **2538** of anchor **2532** at a nonzero angle, due to the chain-link nature of the coupling therebetween. It should be further noted that the use of protrusions **2512** advantageously facilitates the entry of eyelet **2538** into housing **2508** (1) initially at a nonzero angle with respect to the longitudinal axis of the housing, and (2) at a range of rotational positions with respect to the rotational position of the housing, due to the circumferential spaces between the protrusions. Purely for illustrative purposes, this configuration may be contrasted with a hypothetical system in which portions of eyelet **2538** slide into respective slits in the housing, and in which the portions of the eyelet must typically be aligned with the slits before entering the housing.

(126) For some applications, tool **2502** may be used to redeploy tissue anchor **2532** into tissue. For yet other applications, tool **2502** may be used to initially deploy anchor **2532** into tissue. For example, sleeve **228** and/or sleeve **162** described hereinabove may be preloaded with a plurality of anchors **2532**, and tool **2502** may be used to sequentially deploy each anchor into tissue in order to anchor the sleeve to the tissue. For such an application, during the deploying of the plurality of anchors, tool **2502** may also unscrew a given anchor **2532** if the physician determines that the anchor is not positioned correctly.

(127) Typically, but not necessarily, anchor **2532** comprises a biocompatible material such as stainless steel 316 LVM. For some applications, anchor **2532** comprises nitinol. For some applications, anchor **2532** is coated with a non-conductive material.

(128) Reference is now made to FIGS. **16-19**, which are schematic illustrations of a system **1800** for engaging an already-implanted anchor and facilitating extraction (e.g., retrieval) of the anchor from tissue, in accordance with some applications of the invention. System **1800** comprises an anchor-manipulation tool **1802** comprising a tissue-anchor-engaging element **1803** comprising a distal tapered, cone-shaped anchor-engaging tip **1808**, a plunger **1820**, and one or more engaging structures **1810** disposed adjacently to plunger **1820**. System **2500** is typically used in a similar manner to that in which system **100** is used. For some applications, system **1800** is used as described for system **100** with reference to FIG. **10**, *mutatis mutandis*. For some applications, system **1800** is used with techniques described in US Provisional Application 61/717,303 and/or in an International Patent Application entitled, "Controlled steering functionality for implant-delivery tool," to Sheps et al., filed on even date herewith, which was assigned International Application Number PCT/IL2013/050860, and which published as WO 2014/064694, both of which are assigned to the assignee of the present application and are incorporated herein by reference. Two engaging structures **1810** are shown by way of illustration and not limitation. Any suitable number of structures **1810** may be used (e.g., 1-4 structures). FIG. **16** shows an exploded view of tool **1802** in order to illustrate the relationship among the components of tool **1802**. FIG. **17** shows tool **1802** in an assembled state.

(129) As shown in FIGS. **18A-C** in the cross-sectional images, each structure **1810** comprises a curved arm **1811** having an inner surface **1815** which defines (1) a proximal portion **1813** which curves toward a longitudinal axis **1807** of tool **1802** (and toward the other arm), and (2) a central curved portion **1814** which curves away from axis **1807**. Each arm **1811** is shaped so as to define and comprises a distal hook **1816** that is moveable in response to movement of arm **1811**. Arm **1811** is moveable in response to proximal and distal movement of plunger **1820** along axis **1807**. Thereby, plunger **1820** functions as an arm actuator (and therefore as an anchor-engaging-element actuator). As shown in FIG. **16**, plunger **1820** comprises (1) a distal tapered force applicator **1821** having a proximal portion that is wider than a distal portion of force applicator **1821**, and (2) a proximal longitudinal element **1822** coupled to the proximal portion of force applicator **1821**.

Longitudinal element **1822** is narrower than the proximal portion of force applicator **1821**.

(130) Arms **1811** are disposed within an anchor-engaging housing **1804**. Housing **1804** is shaped so as to define one or more (e.g., two) slits **1806**. Each slit **1806** facilitates movement of at least a portion of each arm **1811** radially away from axis **1807**. Housing **1804** is shaped so as to define distal tapered anchor-engaging tip **1808** which is configured to engage an engaging head **1862** of a tissue anchor **1832** (FIGS. **18B-C**) by being received within an opening of head **1862**.

(131) As shown in FIG. **18A**, in a resting state of tool **1802**, plunger **1820** is disposed in a proximal position in which the proximal portion of force applicator **1821** pushes against inner surface **1815** at proximal portion **1813** of arm **1811** such that a force is applied to proximal portion **1813** which keeps arm **1811** and thereby distal hooks **1816** in a closed state. In such a manner, when plunger **1820** is disposed in the proximal position, the proximal portion of applicator **1821** applies a force against proximal portion **1813** of arm **1811** in order to push proximal portions **1813** of each arm **1811** radially outward away from axis **1807**, and to thereby keep hooks **1816** close together, as shown in Section A-A. For some applications, each structure **1810** has a proximal-most portion **1812** which is coupled (e.g., by being welded) to a proximal portion of an outer covering **1830**. When plunger **1820** is disposed in the proximal position, the proximal portion of applicator **1821** applies a force against proximal portion **1813** of arm **1811**, the proximal portion of outer covering **1830** applies a counterforce against structure **1810** to further maintain hooks **1816** in the closed state. Additionally, a distal portion of covering **1830** partially covers slits **1806** and applies additional counterforce against proximal portion **1813** of arm **1811**.

(132) Reference is now made to FIG. **18B** which shows tool **1802** being manipulated into engaging head **1862** of tissue anchor **1832**. As shown, anchor-engaging tip **1808** is advanced within an opening of engaging head **1862**. Engaging head **1862** provides a wall shaped to define the opening, which is shaped to define a funnel having a wide proximal end and a narrower distal end. Engaging head **1862** is shaped so as to define an undercut **1868** and one or more recesses **1869** in a vicinity of undercut **1868**. Each recess **1869** is configured to receive a respective hook **1816** of arm **1811** of tool **1802**. For some applications, head **1862** provides up to 15 recesses **1869**.

(133) As shown in Section B-B, plunger **1820** is coupled at a proximal portion thereof to a plunger-manipulating element **1852**. Element **1852** is slidable within a tube **1850**, which is coupled at a distal end thereof to respective proximal-most portions **1812** of structures **1810**. As indicated by the arrow, element **1852** is advanced distally by sliding within tube **1850**, and thereby, element **1852** pushes plunger **1820** distally in a manner in which the proximal portion of force applicator **1821** is pushed beyond proximal portion **1813** of arm **1811** and force applicator **1821** is disposed adjacently to inner surface **1815** of arm **1811** at central curved portion **1814**. As shown in FIGS. **16** and **18B**, plunger **1820** comprises a proximal longitudinal element **1822** coupled to the proximal portion of force applicator **1821**. Longitudinal element **1822** is narrower than the proximal portion of force applicator **1821** such that upon distal advancement of plunger **1820**, and thereby, distal advancement of the proximal portion of force applicator **1821** beyond proximal portion **1813** of arm **1811**, proximal portions **1813** of arms **1811** incline toward longitudinal element **1822** of plunger **1820**. Responsively, curved central portions **1814** move radially away from axis **1807**, arm **1811** moves, and, responsively to movement of arm **1811**, hooks **1816** are distanced from each other by moving radially away from axis **1807**.

(134) Each hook **1816** moves within a respective slit **1806**. As hooks **1816** move, the proximal portion of outer covering **1830** applies a counterforce against structure **1830** to further maintain hooks **1816** in the closed state. Additionally, a distal portion of covering **1830** partially covers slits **1806** and applies additional counterforce against proximal portion **1813** of arm **1811**.

(135) Reference is now made to FIGS. **18B** and **19**. As shown in Section B-B of FIG. **18B** and in FIG. **19**, at least one hook **1816** engages engaging head **1862** of anchor **1832**, e.g., by hooking around a portion of the head, such as undercut **1868**. In such a configuration, hooks **1816** have not yet been spread apart as far as possible, since force applicator **1821** of plunger **1820** has not yet

been moved as far distally as possible. Thus, hooks **1816** are flexible enough to move radially inward and outward in a manner in which tool **1802** may be angled by the operating physician to optimally engage hooks **1816** with recesses **1869** in the vicinity of undercut **1868**. As shown, tool **1802** is disposed at a nonzero angle with respect to engaging head **1862** of anchor **1832**. Thus, the funnel shape of wall **1866** of engaging head **1862** and the cone-shape of anchor-engaging tip, together with hooks **1816** in their non-fully-deployed state (e.g., intermediate position), enable engaging of tool **1802** with engaging head **1862** of anchor **1832** at an angle α_B (section A-A of FIG. **19**) which is between 0 and 45 degrees, e.g., between 20 and 30 degrees. Angle α_B is the angle between (1) longitudinal axis **1807** of tool **1802**, and (2) a longitudinal axis **1867** of tissue anchor **1832**.

(136) In such a configuration of hooks **1816** with respect to undercut **1868** in FIG. **18B**, tool **1802** is capable of angular movement in order to optimize engaging of hooks **1816** with undercut **1868**. Once the physician is satisfied with the engaging of hooks **1816** with undercut **1868**, the physician pushes down on plunger **1820** by pushing plunger-manipulating element **1852**.

(137) FIG. **18C** shows additional radial movement of hooks **1816** in response to additional distal advancement of plunger **1820**. Upon further distal advancement, the distal portion of force applicator **1821** applies a force against inner surface **1815** of central curved portion **1814**. In response, force applicator **1821** pushes additionally on surface **1815** of curved arms **1811** such that hooks **1816** are additionally pushed radially outwardly, such that they firmly engage undercut **1868**, as shown in Section C-C. Plunger **1820** may then be locked into place in order to ensure hooks **1816** remain locked in place under undercut **1868** of engaging head **1862** of anchor **1832**. In such a configuration, axis **1807** of tool **1802** is aligned with the longitudinal axis of anchor **1832**. Once tool **1802** is positioned as shown in FIG. **18C**, tool **1802** firmly engages anchor **1832** and is then rotated in order to unscrew the already-deployed tissue anchor **1832** from the tissue. For some applications, tool **1802** may be used to redeploy tissue anchor **1832** into tissue.

(138) For yet other applications, tool **1802** may be used to initially deploy anchor **1832** into tissue. For example, sleeve **228** and/or sleeve **162**, described hereinabove, may be preloaded with a plurality of anchors **1832**, and tool **1802** may be used to sequentially deploy each anchor **1832** into tissue in order to anchor the sleeve to tissue. For such an application, during the deploying of the plurality of anchors, tool **1802** may also unscrew a given anchor **1832** if the physician determines that the anchor is not positioned correctly.

(139) Typically, but not necessarily, anchor **1832** comprises a biocompatible material such as stainless steel 316 LVM. For some applications, anchor **1832** comprises nitinol. For some applications, anchor **1832** is coated with a non-conductive material.

(140) Reference is now made to FIGS. **17** and **19**. Typically (as shown), but not necessarily, tool **1802** comprises an annular (or tubular) element **1840** which is slidable around tube **1850** and comprises an annular-positioning guide **1842**. Guide **1842**, as shown, comprises a plurality of radially-branched tines **1844**. Collectively, tines **1844** typically define a circumference having a diameter of between 4 and 5 mm (e.g., 4.5 mm) which is typically at least 1.2 times (e.g., 1.5 times) greater than a largest cross-sectional diameter of tool **1802**. FIG. **19** shows the guided positioning of tool **1802** using guide **1842** of annular element **1840**. Tines **1844** are configured to surround engaging head **1862** of anchor **1832** in a manner which guides tool **1802** and centralizes (e.g., aligns) anchor-engaging tip **1808** within the opening of engaging head **1862**. As shown in the left-most cross-sectional image of Section A-A of FIG. **19**, when tool **1802** initially engages engaging head **1862**, annular element **1840** is slid distally in order to surround at least a portion of engaging head **1862**. Once tool **1802** has initially engaged engaging head **1862**, as shown in the middle cross-sectional image of Section A-A, annular element **1840** may be slid proximally away from engaging head **1862** of anchor **1832**.

(141) System **1800** is typically used in a similar manner to that in which system **100** and/or system **2500** is used. For some applications, system **1800** is used as described for system **100** with

reference to FIG. 10, *mutatis mutandis*. For some applications, annular element **1840** is used to increase a working diameter for tool **1802** by opening a portion of the sleeve **228** and/or sleeve **162**, e.g., a kinked portion of the sleeve. Additionally, and as described hereinabove, in response to the distal pushing of annular element **1840**, annular-positioning guide **1842** helps centralize (e.g., align) anchor-engaging tip **1808** within engaging head **1862** of anchor **1832**.

(142) Reference is made to FIGS. 20A-C, which are schematic illustrations of a technique for use with a tool **80** and anchor **40**, in accordance with some applications of the invention. As described hereinabove, tools **80** and **380** are configured to be articulatably coupled to anchor **40**, to subsequently inhibit this articulation, and to subsequently de-anchor and remove the anchor. For example, FIGS. 4E and 11E show anchor **40** being withdrawn proximally by tool **80** and tool **380**, respectively, while this articulation is inhibited (e.g., in a rigidly-coupled state). FIGS. 10A-C utilize tool **80** as an illustrative example, but the same techniques are applicable to tool **380**, *mutatis mutandis*.

(143) FIG. 20A shows eyelet **46** of anchor **40** disposed within housing **82** of tool **80**, while arms **92** are in the retracted position thereof. Thereby, housing **82** inhibits articulation between tool **80** and anchor **40** (e.g., the tool and anchor are rigidly coupled to each other). In this state, a rigid length **d10** is defined between (1) the distal end of tissue-engaging element **42** of anchor **40**, and (2) a part of tool **80** (e.g., a part of housing **82**) that is proximal to crest **43** of eyelet **46** of the anchor. Typically, rigid length **d10** is defined between the distal end of tissue-engaging element **42**, and a part of tool **80** that is proximal to most or all of arms **92**, such as between the distal end of the tissue-engaging element and a proximal end of housing **82**.

(144) FIG. 20B shows eyelet **46** of anchor **40** disposed outside of housing **82** of tool **80**, while arms **92** are in the intermediate position thereof, i.e., in the articulatably-coupled state described hereinabove. In this state, a rigid length **d11** is defined between (1) the distal end of tissue-engaging element **42** of anchor **40**, and (2) crest **43** of eyelet **46** of the anchor. Typically, a second rigid length **d12** is defined between distal end **83** of housing **82**, and a part of tool **80** that is proximal to most or all of arms **92**, such as between the distal end of the housing and a proximal end of the housing. For some applications, rigid length **d12** is defined between a distal end of arms **92** and the part of tool **80** that is proximal to most or all of arms **92**. Typically, both rigid length **d11** and rigid length **d12** are smaller than rigid length **d10**.

(145) It is hypothesized that, for some applications, it is advantageous to move tool **80** and anchor **40** through at least some parts of a catheter (e.g., catheter **214**) while in the articulatably-coupled state. For example, and as shown in FIG. 20C, because this articulation provides shorter rigid lengths (as described with reference to FIGS. 20A-B), housing **82** and anchor **40** are able to traverse more greatly-bent portions of catheter **14** than they would in the rigidly-coupled state. Transcatheter movement of tool **80** and anchor **40** while articulatably coupled to each other may be used to facilitate both delivery and retrieval of anchors.

(146) For some applications, tool **80** may be configured to maintain the tool and anchor **40** in the articulatably-coupled state (e.g., to prevent rigid coupling therebetween). For example, tool **80** may comprise a stopper, configured to prevent eyelet **46** from entering (e.g., fully entering) compartment **86**. Alternatively, housing **82** may be configured (e.g., shaped) not to receive eyelet **46** (e.g., the housing may be configured to receive only arms **92** and not eyelet **46**).

(147) Reference is again made to FIGS. 2A-7, and reference is also made to FIGS. 21A-C, which are schematic illustrations of a tool **680** for use with tissue anchor **40**, in accordance with some applications of the invention. As described hereinabove, for some applications, only one of the arms of tool **80** defines a concavity. Such a configuration is illustrated by tool **680**. Typically, tool **680** and components thereof are identical to tool **80** and identically-named components thereof, except where noted. Similarly, techniques described herein for use with tool **80** may typically be used with tool **680**, *mutatis mutandis*. Tool **680** is typically configured, and used, to retrieve anchor **40** from tissue of the subject (e.g., after the anchor has been previously implanted and/or used to

facilitate implantation of an implant). For some applications, tool **680** is also configured and/or used to implant anchor **40**.

(148) Tool **680** comprises a housing **682** that has an inner surface that defines a generally cylindrical compartment **686** that has an opening **688** at a distal end **683** of the housing. Tool **80** further comprises an anchor-engaging element **691** comprising one or more (e.g., two) arms **692** (e.g., a first arm **692a** and a second arm **692b**) that define a space **696** therebetween. At least one arm is shaped (e.g., curved) to define a concavity **703**. For example, and as shown in FIGS. **21A-C**, first arm **692a** may be shaped to define concavity **703**. For some applications, arm **692a** comprises arm **92** described hereinabove. Arm **692b** is not shaped to define a concavity, and is typically generally straight. For some applications, a distal portion **694b** of arm **692b** deflects outward from a longitudinal axis of the tool, e.g., so as to facilitate movement of the arm around bar **44** of eyelet **46** of anchor **40**.

(149) Movement of housing **682** with respect to arms **692** moves the arms between an extended position (FIG. **21A**), an intermediate position (FIG. **21B**), and a retracted position (FIG. **21C**), e.g., as described for tool **80** with reference to FIGS. **3A-C**, mutatis mutandis. FIGS. **21A-C** show stages in the transition of arms **692** from the extended position (FIG. **21A**) to the retracted position (FIG. **21C**). FIG. **21B** illustrates that as the arms move from the extended position to the retracted position via the intermediate position, the arms form a loop before assuming the retracted position (e.g., before the completely enter compartment **86**), e.g., as described for tool **80** with respect to FIGS. **3A-C**, mutatis mutandis. In the intermediate position, the arms form a loop that shapes space **696** to define an aperture **697**, e.g., as described for tool **80** with respect to FIGS. **3A-C**, mutatis mutandis.

(150) It will be appreciated by persons skilled in the art that the present invention is not limited to what has been particularly shown and described hereinabove. Rather, the scope of the present invention includes both combinations and subcombinations of the various features described hereinabove, as well as variations and modifications thereof that are not in the prior art, which would occur to persons skilled in the art upon reading the foregoing description.

Claims

1. A system for use with a tissue of a subject, the system comprising: a tissue anchor, comprising: an engaging head, comprising a coupling eyelet that defines an eyelet plane, and a tissue-coupling element that defines a helix that is reversibly screwable into the tissue, and that has a central helix axis that lies on the eyelet plane; and an anchor driver, having a distal portion that is transluminally advanceable toward the tissue, and comprising: a casing that defines a slot, the slot: defining a slot plane, and having a fixed dimension for receiving at least part of the eyelet therein in a manner in which the eyelet plane is coplanar with the slot plane; a detent, actuatable to move with respect to the casing to become disposed through the eyelet in the slot in a manner that reversibly inhibits exiting of the eyelet from the slot; and the anchor driver being configured to screw the tissue anchor into the tissue by rotating the tissue anchor about the central helix axis by applying a rotational force to the eyelet.
2. The system according to claim 1, wherein the slot is dimensioned to receive the at least part of the eyelet snugly.
3. The system according to claim 1, wherein the anchor driver further comprises a controller, configured to move the detent between (i) a closed state in which the at least part of the eyelet is inhibited from exiting the slot, and (ii) an open state in which the at least part of the eyelet is slidable out of the slot.
4. A system according to claim 3, wherein: the casing defines a recess, at least part of the detent moves into the recess when the detent moves into the open state thereof, and the controller comprises a rod, configured to move the detent into the closed state thereof by the rod being slid

into at least part of the recess.

5. The system according to claim 1, further comprising an anchor-retrieval tool, transluminally advanceable independently of the anchor driver, wherein the anchor-retrieval tool comprises: an anchor-engaging element comprising one or more arms, configured to be coupled to the engaging head, a housing that defines a compartment therein, and a retrieval-tool controller, comprising an adjuster configured to move at least part of the anchor-engaging element into and out of the compartment.

6. The system according to claim 1, further comprising an anchor-retrieval tool, transluminally advanceable independently of the anchor driver, wherein the anchor-retrieval tool: comprises, at a distal portion thereof, an anchor-engaging element comprising a curved arm, configured to be coupled to the tissue anchor by moving through the eyelet, and is configured, while the anchor-engaging element is coupled to the tissue anchor, to apply a rotational de-anchoring force to the tissue anchor.

7. The system according to claim 6, wherein the curved arm is generally helical, and is configured to be coupled to the eyelet by being rotated with respect to the eyelet.

8. The system according to claim 7, wherein the tissue-coupling element has a handedness, and the curved arm has a handedness that is opposite to the handedness of the tissue-coupling element.

9. The system according to claim 6, wherein the anchor-engaging element is configured to be coupled to the tissue anchor by being rotated with respect to the tissue anchor, and the anchor-retrieval tool is configured to apply the rotational de-anchoring force to the tissue anchor by at least part of the anchor-engaging element being rotated with respect to the tissue anchor.

10. The system according to claim 1, wherein the detent is configured to transition between (i) a closed state in which the at least part of the eyelet is inhibited from exiting the slot, and (ii) an open state in which the at least part of the eyelet is slidable out of the slot.

11. The system according to claim 1, wherein: the casing further defines a recess in communication with the slot, and the detent is actuatable to move with respect to the casing to become disposed through the eyelet in the slot by being actuatable to move out of the recess and into the slot to become disposed through the eyelet in the slot.

12. A system for use with a tissue of a subject, the system comprising: (i) a tissue anchor, comprising: an engaging head, comprising a coupling eyelet that defines an eyelet plane, and a helical tissue-coupling element, reversibly anchorable to the tissue, and having a central helix axis that lies on the eyelet plane; (ii) an anchor driver, configured to anchor the tissue anchor to the tissue by rotating the tissue anchor about the central helix axis by applying a rotational force to the eyelet, the anchor driver comprising: a casing that defines a slot, the slot defining a slot plane, and dimensioned to receive at least part of the eyelet therein in a manner in which the eyelet plane is coplanar with the slot plane, and a detent, configured to reversibly inhibit movement of the eyelet with respect to the slot; and (iii) an anchor-retrieval tool, transluminally advanceable independently of the anchor driver, the anchor-retrieval tool comprising: at a distal portion, an anchor-engaging-element actuator, and an anchor-engaging element comprising a curved arm, and configured to be coupled to the tissue anchor by moving through the eyelet, and at a proximal portion of the anchor-retrieval tool, a retrieval-tool controller configured to control the anchor-engaging-element actuator, wherein: the system has at least two states: (1) an uncoupled state in which the anchor-engaging element is not in contact with the tissue anchor, and (2) an articulatably-coupled state in which: the engaging head at least in part inhibits movement of the anchor-engaging element away from the tissue anchor, and the distal portion is deflectable with respect to the tissue anchor; the anchor-engaging-element actuator is configured to facilitate transition of the system from the uncoupled state to the articulatably-coupled state; and in at least one of the at least two states, the anchor-retrieval tool is configured, while coupled to the tissue anchor, to apply a rotational de-anchoring force to the tissue anchor.

13. The system according to claim 12, wherein in the articulatably-coupled state, the distal portion

of the anchor-retrieval tool is rotatable with respect to the tissue anchor.

14. The system according to claim 13, wherein the anchor-retrieval tool is configured to rotate the tissue anchor by applying the de-anchoring force to the tissue anchor.

15. The system according to claim 12, wherein the at least two states further comprise a rigidly-coupled state in which the distal portion of the anchor-retrieval tool is inhibited from articulating with respect to the tissue anchor.

16. The system according to claim 15, wherein the anchor-retrieval tool is configured to apply the de-anchoring force to the tissue anchor while the system is in the rigidly-coupled state.

17. The system according to claim 15, wherein the anchor-engaging-element actuator is configured to transition the system from the articulatably-coupled state to the rigidly-coupled state by sliding, with respect to the anchor-engaging element, along a longitudinal axis of the distal portion of the tool.

18. The system according to claim 12, wherein the anchor-engaging element comprises at least a first arm and a second arm, a curve of each of the arms defining a respective concavity, and the arms positioned such that the concavity of the first arm faces the concavity of the second arm.

19. The system according to claim 18, wherein the anchor-engaging element further comprises a housing that defines a compartment therein, and an adjuster that is configured to move at least part of each of the arms into and out of the compartment.

20. The system according to claim 19, wherein, when the anchor-engaging element is coupled to the tissue anchor, the anchor-retrieval tool is configured to move at least part of the eyelet into the compartment by the at least part of each of the arms moving into the compartment, and the housing is configured to apply the rotational de-anchoring force to the tissue anchor by the housing being rotated while the at least part of the eyelet is disposed within the compartment.
